Participation Costs, Search Time, and Match Quality in the Housing Choice Voucher Program*

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Abstract

Many social safety net programs struggle with incomplete program take-up, oftentimes due to high participation costs. The Housing Choice Voucher Program, the largest rental assistance program in the country, involves a complex, multi-step take-up process and coordination with the private rental housing market, creating considerable barriers to entry for participants. In this paper, I evaluate the effect of time constraints in the voucher take-up process on program participation and the quality of the housing match for successful voucher participants. Specifically, I look at two unique features of voucher take-up that cause time constraints: the random timing of voucher issuance within a month, and variation in how long voucher recipients are given to search. In the aggregate, receiving a voucher at the end of a calendar month reduces the likelihood of using that voucher within certain time frames, but has no effect on the probability of a participant ever using their voucher. However, housing authorities vary in their time restrictions on voucher use and their leniency in granting extensions. At housing authorities that do not offer extensions, end-of-month issuance results in a 6 percentage point decline in voucher success rates. Households at these housing authorities see shorter search times, and as a result, are more likely to end up in over-crowded housing situations. These results illustrate how flexible search time policies can alleviate some of the high participation costs of voucher use.

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1 Introduction

Social programs often struggle with incomplete take-up. In many cases, this can come from difficulties in learning about programs, documenting eligibility, and meeting all necessary requirements (Herd and Moynihan, 2018). The Housing Choice Voucher Program, the largest federal rental assistance program in the country, is intended to improve affordability, reduce the prevalence of unsafe or overcrowded housing conditions, and expand neighborhood choice for low-income households. However, the voucher program involves heightened challenges for participation. In addition to completing the program application steps, participants in the voucher program must also find and rent eligible homes on the private rental market in order to receive their subsidies. This means that the already challenging private market housing search process is inserted into the program take-up equation. In recent years, the tightening of rental markets across the country has intensified the difficulty of using vouchers; in 2022, only 57% of households who were issued vouchers were able to use them.

In this paper, I study two features of the voucher program take-up process that impose constraints on search times for participants, which can in turn affect who is able to receive assistance. Search time, or the amount of time between voucher issuance and a household leasing a home with that voucher, is a crucial metric for the voucher program. This is in part because potential voucher recipients are only given a set period of time to locate a home with their voucher, after which they must return the voucher and will not receive any assistance. Search time can also illustrate the difficulty of the take-up process, even for households who do ultimately use their vouchers. In recent years, search times for successful households have increased sharply as households contend with rapidly rising rents, a shortage of affordable housing units, and the difficulty of finding landlords who will participate in the program.

The first feature of the program that can create time constraints for voucher recipients is the random timing of voucher issuance. While households may receive a voucher on any day of the month, leases in the private rental market are most likely to start on the first of the month. The day of the month that a voucher is issued could affect voucher take-up since households who are issued a voucher at the beginning of the month may have more effective search time and are therefore more likely to successfully find and lease a home in the private market before the voucher expires.

The date of issuance could also affect housing match quality for households who do still manage to use their vouchers since households who are issued vouchers at the end of the month have less time to find a housing unit that meets their needs and preferences.

The second feature is that housing authorities directly impose time constraints by limiting how long households can search for housing after receiving their vouchers. The voucher program is administered by more than 2,000 public housing authorities (PHAs) across the country, and PHAs have a substantial amount of discretion over the operations of the program, including setting allowable search times. While federal regulations require that PHAs give households at least 60 days to search, there is no maximum amount of search time. In practice, many PHAs allow for search times above and beyond federal requirements through longer initial search times or extension policies. However, allowable search time policies can be endogenous, as PHAs may set more generous search time policies where market conditions or other factors make it more difficult to households to use their vouchers. This makes it difficult to directly evaluate the effects of allowable search time policies on the ability of households to use their vouchers, and other outcomes such as the neighborhoods where they are able to live.

To understand the effects of these time constraints on voucher program participation and housing match quality, I leverage the random timing of voucher issuance to generate causal estimates on how end-of-month issuance affects search outcomes, and how stringent search time policies can interact with issuance timing to impose substantial costs on voucher recipients. I use two novel data sources to measure each of these constraints and how they interact with each other. First, I employ confidential administrative data on success rates and search times for nearly 1,000 PHAs. While past studies of success rates have relied on direct collaboration with a small set of PHAs (Finkel and Buron, 2001; Chyn et al., 2019), using administrative data vastly expands the ability to measure these key outcomes. Second, despite the importance of allowable search time policies in the take-up process, previously there has been no centralized source of information on these policies. I collect more than 800 PHA administrative plans to construct the first consolidated source of information on allowable search time policies, covering 433 unique PHAs.

Focusing first on the direct effects of issuance timing within the month, I find that new voucher recipients who are issued a voucher at the end of the month are less likely to use their voucher within 60, 90, and 120 days. These effects have persisted over time, and as search times have grown, the effects of end-of-month issuance are observed at even longer search windows. However, in the aggregate, there is no effect on the ability of households to ever use their voucher. This can be attributed, at least in part, to the flexibility that PHAs have to extend voucher terms and suspend the term of the voucher while housing units undergo inspections.

I conduct heterogeneity tests to see if issuance timing may affect targeting within the program; in other words, if certain groups are more affected by the timing of their voucher issuance, and may lose access to assistance as a result. In general, most racial/ethnic and household composition groups follow the same pattern as the program as a whole, where they are less likely to use their vouchers within certain time periods, but not less likely to ever use their vouchers; however, I find that households with a household head over the age of 65 are about 1% less likely to ever use their voucher when they are issued a voucher at the end of the month.

Given the decline in success rates at shorter search windows but a lack of overall effects on program participation, the timing of voucher issuance has two potential implications: first, some households who still succeed in using their vouchers will have had less time to search for housing, which may affect their housing match quality; and second, households who are unable to use their vouchers within certain search windows may lose access to assistance altogether if PHAs enforce more stringent time constraints.

I test the first hypothesis and find that, in the aggregate, there are no meaningful effects on housing match quality for those who are successful in using their vouchers, measured as the ability to reach lower poverty neighborhoods, the length of time that a household remains in a housing unit, and the relative size of the unit that a household leases. There are, however, important differences in the effects of end-of-month issuance with regard to PHA search time policies. With the data I collected on allowable search times, I provide the first descriptive information on PHA search time policies, through which I identify variation in initial search terms and extension policies that either directly or effectively extend allowable search times.

I use this variation to test the second hypothesis, that issuance timing can cause households to miss certain deadlines that could lead to the household losing their voucher altogether if PHAs impose more stringent time constraints. Although many PHAs do give the minimum mandated 60-day voucher term, a majority of PHAs allow for some type of extension, with this practice becoming more common over time. However, there remains a small subset of PHAs that only allow the minimum search time and in general do not provide any extensions to non-disabled households. I find that at PHAs with the strictest search time policies–a 60-day initial term with no extensions–recipients who receive their voucher at the end of the month are 6 percentage points less likely to ever use their vouchers. This illustrates how, without the flexibility afforded by more lenient extension policies, the loss of effective search time can cause declines in take-up, even for households who are eligible and have repeatedly expressed interest in the program. This effect is strongest for households with children, and persists across racial groups.

There is also evidence of differential effects on match quality, driven by PHAs that do not allow extensions, where end-of-month issuance is most likely to reduce effective search time. More stringent PHAs see substantial declines in search time, with those who receive their vouchers at the end the month on average searching for 8 fewer days than those issued a voucher in the first three weeks of the month. At this set of PHAs, successful households are more likely to lease up into smaller units than they qualify for, compared to households who receive their voucher earlier in the month. This is a key finding, as one of the core goals of the voucher program is to reduce overcrowding.

This paper contributes to several strains of literature. First, I contribute to the literature on administrative burden and barriers to program participation. Most closely related is Homonoff and Somerville (2021), which documents how the timing of re-certification interviews within a month can affect SNAP participation. More broadly, an existing strain of literature examines how high participation costs can affect program take-up (Currie, 2006; Chetty et al., 2013; Bettinger et al., 2012; Deshpande and Li, 2019). This is the first paper to adopt this framework to housing assistance, enhancing our understanding of participation costs in the context of a program that requires interaction with the housing market and a private third party.

Additionally, I contribute to a growing literature on the operations of the voucher program, and the intersection between program take-up and housing search. This paper is one of the first in a small but growing set of papers to estimate voucher success rates (Ellen et al., 2024; Chyn et al., 2019; Finkel and Buron, 2001; Reina and Winter, 2019; Shroder, 2002). My results on housing match quality add to a body of work that studies how the voucher program and other mobility programs can better meet their goals of encouraging neighborhood choice (Bergman et al., 2024; DeLuca et al., 2023; Ellen et al., 2025). I also add to the literature on discretion in the voucher program, and how local housing authorities use their discretion in administration practices that can shape outcomes for participants (Zhang and Johnson, 2023; McCabe and Moore, 2021; Hembre and Urban, 2023; Denton et al., 2014; Dawkins and Jeon, 2017; McCabe, 2023). Finally, I speak to the literature on the low-income housing search process more broadly, providing new evidence on how time constraints can affect the ability of low-income households to find housing that meets their needs and preferences (Harvey et al., 2020; Krysan and Crowder, 2017).

The paper proceeds as follows. Section 2 lays out the institutional context, including detailing the housing voucher take-up process. Section 3 describes the administrative data used to calculate success rates and the originally collected data on PHA search policies. Section 4 presents descriptive facts on success rates, search times, and PHA search time policies. Section 5 shows results on the effects of issuance timing on success rates, and section 6 shows the effects of issuance timing on measures of match quality. Section 7 presents heterogeneity in results on success rates and match quality by PHA search time policies, and section 8 discusses the implications of these findings.

2 Institutional context: Housing Choice Voucher program

The Housing Choice Voucher program is the largest federal rental assistance program in the country. In 2022, more than 5 million individuals in 2.3 million households received vouchers (Center on Budget and Policy Priorities, 2022). While the program is funded federally by the Department of Housing and Urban Development (HUD), it is administered locally by more then 2,200 public housing authorities (PHAs). PHAs have discretion over the daily operation of the program, but abide by certain federal guidelines set by HUD.

Vouchers can be transformative for low-income households who are able to use them. Once a household enters the voucher program, they generally pay 30 percent of their income towards rent, and the local housing authority covers the rest, up to a certain subsidy limit, called the payment standard. Rent burdens in the voucher program are significantly lower than those of unassisted, low-income renters (Ellen and Torrats-Espinosa, 2020). Vouchers have also been shown to reduce homelessness, reduce over-crowding, and improve children's performance in school (Jacob and Ludwig, 2012; Mills et al., 2006; Wood et al., 2008; Schwartz et al., 2020).

However, the program struggles with two main issues. This first is that many households who receive vouchers are not able to use them. In recent years, up to 40 percent of households who receive vouchers must return them to the housing authority as they are unable to find and rent a qualified home within the allotted time they are given. Second, the voucher program has not consistently achieved a core goal of allowing voucher holders to live in a diverse set of neighborhoods (Galvez, 2011; Pendall, 2000). Both of these issues can reflect the difficulty that voucher recipients may have during the search process, and the high participation costs that the program entails.

2.1 Take up in the voucher program

Take-up in the voucher program is a multi-step process, often spanning multiple years. First, voucher holders must apply to join a wait list at a PHA. Many PHAs will close their wait lists due to out-sized demand for the program, so in order to even apply, participants must often wait for the PHA in their local market to open their wait list[1]. At this point, households apply, and if they are certified to qualify for the program, they are placed on the wait list. While wait times vary widely across the country, they are generally quite long. On average, households wait 2.5 years, but at some housing authorities included in the sample for this paper, average wait times can be as long as 8 years (Acosta and Gartland, 2021).

Once a household is selected off the wait list, they are contacted by the PHA. At this point, they confirm their eligibility and interest in the program, and provide documentation. Only then are they issued a search voucher. A search voucher is a promise of receiving the subsidy, given the voucher recipient is able to secure a qualifying unit. At this point, HUD mandates that households must be given a minimum of 60 days to search, though many PHAs offer longer search windows. PHAs also have discretion to set their own policies around extensions. Many PHAs offer some sort of automatic extensions, and virtually all PHAs grant extensions for a wide variety of acceptable reasons. There is no limit on the number of extensions that PHAs can give, and PHAs are encouraged to consider factors such as market conditions, the approximate wait time for households on the wait list, and the likelihood that the household will be successful if they are given more time (HCV Program Guidebook, Chapter 2). All PHAs must give extended time as a reasonable accommodation for households with disabilities.

[1] See Moore (2016) for a discussion of wait list management strategies; Hembre and Urban (2023) find that between 2010 and 2017, just over 50% of the 1,000+ PHAs they collected data for had their wait list closed for at least part of the time period, and 4% of PHAs never opened their wait list during that time.

Households now must find a unit that meets all programmatic requirements. The unit must have a rent that falls within the payment standard, and must pass a housing quality inspection.[2] The landlord for the unit must also be willing to rent to a voucher holder and participate in the program; once a household leases up with their voucher, the housing authority pays the government portion of the rent directly to the landlord.

Finding a home that meets these requirements and a landlord that will agree to participate in the program is a high hurdle. There is a large body of evidence that shows that landlords discriminate against voucher holders, and are perhaps even more likely to do so in higher opportunity neighborhoods (Phillips, 2017; Blanco and Song, 2024; Cunningham et al., 2018). There is also evidence that households with children and Black and Hispanic households, which constitute the majority of voucher holders, may face heightened discrimination as well, which can affect their ability to use their vouchers, and the locations where they are able to live even when they are successful (Faber and Mercier, 2022; Briggs et al., 2010; Shroder, 2002).

Vouchers are not an entitlement, which makes calculating a traditional take-up rate difficult. Only one in five low-income households with unmet housing needs gets any form of housing assistance (Kingsley, 2017). In this paper, I focus only on eligible households who are issued a voucher, a small subset of all eligible households. The share of households who are able to use their vouchers once they are issued a voucher is referred to as the success rate; a search is generally considered successful if the household is able to use the voucher within one year of issuance.[3] It's important to note that at this point in the process, households have repeatedly expressed interest in the program and confirmed their eligibility. The voucher program also provides quite generous subsidies. The average household in the sample for this paper received $8,000 per year, and in more expensive housing markets, this number can be substantially higher. This makes low, and declining, success rates even more striking, and invites further examination of voucher take-up process.

[2]Payment standards are based on HUD's Fair Market Rents (FMRs). PHAs have discretion to set payment standards within 90-110% of FMRs. Voucher holders may rent units above the payment standard, but they are required to pay the difference between the rent of the unit and the payment standard out-of-pocket, and when a voucher recipient first enters the program, their rent payment cannot exceed 40% of their income.

[3]The one year window allows for comparable measures across years. Most successful searches will conclude within a year; in the sample for this paper, 99% of searches that ever succeeded concluded within one year.

2.2 Time constraints in the take-up process

Part of the reason that success rates remain low in the voucher program is that the take-up process requires participants to navigate the private housing market search. In recent years, this has become an increasingly difficult challenge for all low-income households, as rents have risen significantly and a lack of affordable rental homes persists across the country.

Voucher recipients contend with these same challenges as unassisted households, but also have to navigate all the requirements of the program described above, and must do so within a limited time frame, which is dictated by the housing authority where the household receives their voucher. PHAs have the discretion to set their own search time policies, given that they provide at least a minimum of 60 days. When setting search time policies, PHAs are balancing different priorities, including success rates for participants, utilization rates (or the share of their budget that is leased up into vouchers), and the speed at which they are issuing vouchers and clearing their wait lists. PHAs also have to contend with staff capacity constraints; PHA staff need to hold briefings, grant extensions, conduct inspections, and in some cases, provide assistance to voucher recipients in the search process.

An additional friction for voucher recipients is that while they may receive their voucher on any day of the month, in general, leases for homes start on the first of the month. This can put voucher recipients who are issued their vouchers toward the end of a month at a disadvantage if they are given, for example, only 60 days to use their voucher. Households who are issued vouchers at the end of month will have fewer days of effective search time if they intend to start their lease within 60 days of their voucher being issued, given they are most likely to find leases that start on the first of the month.

If PHAs strictly enforce initial search terms, this could lead to lower success rates for households who start their searches at the end of the month. However, if PHAs offer extensions, or longer initial search terms, we may see only a delay in when households are able to use their vouchers and start receiving benefits, and no change in the overall success rates. Therefore, the ultimate effect of issuance timing can depend on how PHAs set their search time policies, and how they choose to implement certain features of their own policies, such as leniency around issuing extensions.

2.3 Housing match quality

There also could be potential impacts of PHA time limits and issuance timing on the quality of the match between households and the housing units that they find. Research has shown that low-income households may struggle to find housing in shorter periods of time (DeLuca et al., 2013). Parallel to the literature on job search that generally finds households find a better employment match when given longer to search (Van Hooft et al., 2021; Caliendo et al., 2013; Gaure et al., 2012; Belzil, 2001), if households are dealing with a more constrained search window, they may accept housing units that they would otherwise decline, had they been given more time to search. Both the time limits that PHAs set and the timing of voucher issuance can therefore affect match quality.

When PHAs set longer time limits, voucher recipients may approach the housing search differently, compared to at PHAs with shorter initial search times, where households may be most concerned about using the voucher before it expires. This could be particularly important in the voucher program, where households often have difficulty in using their voucher to move to more expensive neighborhoods, where both program rent ceilings and landlord reluctance to house voucher holders may make it more difficult or time-consuming to use a voucher (Ellen et al., 2025; Phillips, 2017; Blanco and Song, 2024).

The timing of voucher issuance may also affect match quality in a similar way, in that if households who are issued a voucher at the end of a month have less time to search, they may not find units that meet all of their goals. However, some households will take longer to search if their issuance was at the end of the month, because they will roll over and take an additional month to find a housing unit, given that their PHA grants them this flexibility. In this case, we may expect to see positive impacts on housing match quality. PHA search time policies also likely shape the effects of issuance timing on match quality, as at more stringent PHAs that do not offer extensions, end-of-month issuance may lead to a more stark loss in search time.

3 Data

3.1 Administrative data on program participation

I use detailed administrative data from the Office of Public and Indian Housing (PIH), the division of HUD that is responsible for administering the voucher program. PHAs submit data on each programmatic action to the PIH Information Center (PIC).[4] These actions indicate when each household is issued a voucher, and if and when they are successful in using their voucher. From these actions, I define my key metrics: indicators for a voucher recipient successfully using their voucher, and the elapsed time between the issuance of the voucher and starting a lease with that voucher, for households who are successful. I also create indicators for a household successfully using their voucher within certain key search windows, including 60, 90, 120, 180, and 360 days.

Importantly, I observe key demographics, including race, ethnicity, age, household composition, and origin ZIP Code at the time of voucher issuance. For households who are successful, once they are admitted to the program, I observe their full demographics, complete address, and detailed information on their housing costs and subsidy payments. For this set of households who are successful, I also observe all subsequent actions, including moves, ports to other PHAs, income re-certifications, and program exits. From these additional actions, I can construct a history of program participation, and identify how long households remain in the voucher program and at a specific address.[5]

Another key feature of this data is that I observe the ZIP Code where voucher recipients live when they receive their voucher, and where they live once they lease-up, if successful. From this information, I derive a set of neighborhood outcomes that measure whether or not households are able to move to neighborhoods with lower poverty rates. Specifically, I merge the administrative data with data from the 2015-2019 American Community Survey (ACS) on poverty rates and poverty percentiles within a CBSA at the ZCTA level. I identify a dichotomous indicator for moves where the destination poverty percentile is higher or lower than the origin poverty percentile. Lastly, I use county rent levels from the 1-year 2019 ACS to stratify the sample into median country rent

[4] More detail on this data source can be found in Appendix A.

[5] Specifically, I detect changes in addresses by geocoding each unique address entry. The geocoding match rate is above 95%. I then use the cleaned addresses to detect any changes in address. This methodology ignores moves within the same building, so a move is defined as a change of street address, not simply a change in unit. Changing units within a building is relatively rare; I estimate that only around 3% of moves are within the same building.

quartiles to conduct heterogeneity analyses.

The sample for the administrative data on success rates and search times is expansive, containing 950 PHAs in total. In the years between 2015 and 2022, these PHAs issued over 1.1 million vouchers. I include only PHAs that meet administrative data quality standards as detailed in Ellen et al. (2023), and explained in more detail in Appendix A. I also restrict the sample to include only PHAs that are predominately located in one county, in order to be able to accurately measure market conditions consistently across PHAs. Combined, these PHAs serve around 47% of the total participants in the program nationally, and in general, and very representative of the program as a whole (Appendix table A1).

Table 1 shows important characteristics of the administrative data sample. Around 45% of new voucher recipients are Black, and 14.3% are Hispanic. Additionally, almost three quarters of households are headed by females, with the majority (53%) of these households containing children.

3.2 Collected data on PHA search time policies

There is no centralized source of information on allowable search time policies by PHA, despite HUD detailing the importance of these policies (HUD Voucher Guidebook, Chapter 2). PHAs record these policies in annual Housing Choice Voucher Administrative Plans, which do not receive formal approval from HUD, so records of these plans are not maintained. In total, I collect and analyze 812 administrative plans for 433 unique PHAs that come from several different sources. First, I hand-collected 410 plans from PHA websites and direct outreach to PHAs. These plans generally cover the years 2023 and 2024, though some PHAs either post or were able to provide historical plans that overlap with my administrative data time period. Second, historical plans for 402 PHAs in my administrative data sample were provided by Zhang and Johnson (2023), who hand-collected plans from PHA websites and outreach between 2016 and 2020. These two sources combined paint a picture of both past and present search time policies.

In order to both use the details of the plans for this analysis and create a data resource that will not only illustrate the current state of allowable search time policies, but also enable a future body of work on changes in search time policies, I focused on collecting plans for PHAs that I can observe at multiple points in time. Most (62%) PHAs in my sample are observed at multiple points in time; however, in general, it's likely that I only observe each PHA once during my administrative data time period, between 2015 and 2022. When I merge the data on collected plans with the administrative data on success rates, I include only observations for PHAs in years where I directly observe their search time policies. This results in approximately 90,000 search events across 507 PHA-years, which constitutes 433 unique PHAs. In table 1, I show that the sample of PHAs with allowable search time policies looks largely similar to the main administrative data sample, although households in the allowable search time sample are more likely to be Black or Hispanic.

From the administrative plans, I record data on the initial search term, automatic extensions, discretionary extensions, tolling and expiration practices, and any exceptions that PHAs make in search times. I then also classify PHAs by their "effective" allowable search time, which is their initial search time plus time given from any automatic extensions. More information about the data collection process can be found in Appendix B.

4 Descriptive facts

4.1 Search times and success rates

Low success rates have long been a key challenge for the voucher program, but until recently there was very little insight into voucher success rates or search times. The administrative data used for this paper is one of the only sources of data that allows for measuring both how many households are able to use their vouchers, and how long this process takes, on a representative scale and over time.

Prior to 2021, success rates (defined here as the share of households who are able to use their voucher within one year of issuance) hovered around 65%-68%, but have since declined sharply. By 2022, only 57% of those issued a voucher were able to lease up and enter the program (figure 1, left panel). At the same time, search times have been sharply increasing. In 2022, it took the median voucher recipient around 77 days to use their voucher, up more than two weeks from two years prior (figure 1, right panel). Both of these metrics can indicate growing search costs for voucher recipients.

Another way to measure success rates and to understand the relationship between growing search times and success rates it to look at success rates across different search windows. For instance, a 60-day success rate is defined as the share of voucher recipients who are successful in using their vouchers within 60 days of issuance. This metric is important, considering that voucher recipients are given a set time period to use their voucher; we can use these search windows to see what share of voucher recipients are able to succeed in a given time frame.

Figure 2 shows success rates at 60-, 90-, 120-, 180-, and 360-day search windows, for the years 2015, 2019, and 2022. First, we see that a relatively low share of voucher recipients are able to use their vouchers within 60 days, the federal minimum amount of search time that PHAs are required to give. Even in 2015, when success rates were higher and search times were shorter, only 36% of voucher recipients succeeded in starting leases using their vouchers within 60 days of voucher issuance.

Additionally, it's important to note that success rates declined at every search window between 2015 and 2022, and we see very striking differences when we look across shorter time intervals. For instance, in 2015, around 66% of voucher holders were able to use their voucher within 6 months of receiving it. In 2022, only 50% of voucher recipients were successful within 6 months. This shows that voucher recipients are facing growing search costs, and that a significant portion of voucher recipients are now searching for more than 6 months to be able to use their voucher.

There is also wide variation in success rates and search times across the country (Figure C1). Ellen et al. (2024) explain some of this variation with a combination of individual characteristics of households, characteristics of neighborhoods where searches originate, and market conditions, but a large share of this variation still remains unexplained. Differences in search time are particularly large, which can be due to differences in PHA policies, differences in market conditions, and how PHAs may respond to differences in market conditions.

In fact, market conditions have a particularly strong association with search times. Appendix figure C2 shows median search times by county median rent quartile. Successful searches in the highest rent counties are almost twice as long as in the lowest rent counties in 2021 and 2022. However, this does not equate to higher rent counties experiencing lower success rates. In fact, when we look at success rates over a one year search window, higher rent counties actually have considerably higher success rates than lower rent counties (figure C2, left panel). This runs counter to a common prediction that success rates will be lower where markets are tighter; it can also emphasize the complicated relationship between search times and market conditions, as PHAs may set their search time policies or exercise more flexibility within existing policies as a response to tighter housing markets.

4.2 Allowable search time policies

A question that arises from the stylized facts presented thus far is how much of the variation in search time comes from differences in allowable search time, as opposed to differences across households within a certain set of constraints. The data assembled for this paper contain the first ever source of information on PHA search time policies. This is important for several reasons. First, extending search times has been cited as a potential strategy to boost success rates across the program, and particularly to mitigate racial disparities in success rates (Ellen et al., 2024). Second, information on PHA policies can be used to determine how flexibility in search times can affect a wide variety of outcomes for participants.

Across the 507 administrative plans from 2015-2022, the majority of PHAs give the federal minimum 60 day initial search term (table 2). However, most PHAs allow for extensions, and many PHAs even give automatic extensions.[6] Combined, around 40% of PHAs have a maximum term limit. Maximum term limits serve as a cap on the length of time that a voucher can be extended; these are most common among PHAs with 60 day initial terms. Additionally, nearly all PHAs allow for "tolling", or suspension of the voucher term when a unit is under inspection, which was made mandatory by HUD in 2015. More information about the prevalence of extensions and other characteristics of PHA search time policies and how these characteristics vary over time can be found in Appendix B.

Importantly, however, initial search times are not always closely related to observed search time. PHAs that have a 60-day initial search terms actually tend to have longer searches, on average, than PHAs with 90 or 120 day search times (table 3). This indicates that much of the difference in search time across PHAs is likely due to how common it is for voucher recipients to seek out extensions, and lenient PHAs are in granting such requests. In fact, at PHAs with a 60-day initial search terms, more than half of successful searches succeed more than 15 days past the initial search time.[7] I also construct a measure of a PHA's "effective initial search term". This measure takes

[6]The difference between automatic and discretionary extensions is that in the case of an automatic extensions, generally households must still request the extension, but their request will not be denied. In the case of a discretionary extension, a PHA may deny the request or require the household to meet certain conditions before granting additional search time.

[7]I use 15 days since most PHAs indicate that inspections are to be completed within 15 days; this approximates

the initial search term, and adds the number of days that would be granted from an automatic extension, if the PHA offers automatic extensions. However, we still see that close to half of searches at PHAs with 60-day initial terms succeed 15 days past the effective initial search term as well; this is expected, as all PHAs that offer automatic extensions also allow for additional discretionary extensions as well.

5 Effects of issuance timing on success rates

In this section, I use a series of models to document how issuance timing can lead to a decrease in the probability of a household using their voucher within certain search windows. The timing of voucher issuance can place an additional constraint on voucher recipients who already have to contend with high participation costs and an increasingly difficult housing search. In appendix figures D1 and D2, I show that search vouchers are issued–or in other words, searches can begin–across the days of the month, but searches most commonly conclude on the first of the month, when households are able to start their leases.

Below, I use the following equation to measure the effect of the day of the month that a search voucher is issued on the probability of a household successfully using their voucher within a certain search window. The equation takes the following form:

$$Success_{itdp} = \beta \sum_{j=1}^{31} \mathbb{I}(d - j = 0) + \ \gamma X_i' + PHA_p \times Year_t + \varepsilon \tag{1}$$

where j is an indicator for each day of the month (1-31) and d is the day of issuance for household i in year t and PHA p. Importantly, I use PHA by year fixed effects to hold constant allowable search time policies at this stage. I also control for individual characteristics, including race/ethnicity, age, household composition, disability status, homeless status, and citizenship status.

These models are identified by the fact that the exact timing of voucher issuance is random. Several features of the program's administration support this assumption. First, households tend to wait on wait lists for many years, and since many wait lists are commonly closed, in practice many households likely even apply on the same day. This means that voucher recipients have no ability to influence when they receive their voucher by strategically applying at a certain time. Second,

any tolling time that the household may receive.

even though PHAs may have different prioritization schedules for issuing vouchers, in general these schedules do not line up with the days of the month. Balance tests found in Appendix D support this, showing that there are no significant differences in observed demographics across days of the month.

First, I am using 30-day success rates to emphasize the mechanical relationship between the day of voucher issuance, and when a household is able to lease up (figure 3, top left panel). As the month progresses, it's harder for households to start a lease by the first of the next calendar month, which leads to fewer households who are able to lease up within 30 days of their issuance. By the end of the month, this effect is around 5 percentage points, which is a 38% decrease from the beginning of the month (only around 13% of searches are successful within 30 days).

Next, I show coefficients on indicator variables for the day of the month of voucher issuance for 4 additional search windows: 60-day success, 90-day success, 120-day success, and an unrestricted measure of success rates (Figure 3). The effect of timing within a month persists beyond the relationship with 30-day success rates. Focusing first on the top left panel, we see that search events that begin at the end of the month are significantly less likely to succeed within 60 days of voucher issuance. At the very end of the month, the effect is around 3 percentage points, which is an economically significant effect given only around 30 percent of searches succeed within 60 days, equating to approximately a 10% decline.

The effect of end of month issuance persists when looking at 90 day success rates (top right panel). Searches that start towards the end of the month are 1 to 1.5 percentage points less likely to succeed within 90 days. This is a smaller but still noticeable effect, as around 43% of searches succeed within 90 days.

At a 120 day search window, the effect is much more muted; though coefficients are negative at the end of the month and are much lower than some periods in the middle of the month, there is no significant effect compared to the reference period at the very beginning of the month. And finally, looking at the bottom right panel, there are no clear patterns when looking at an unrestricted measure of success rates. In other words, end of month issuance has a significant effect on the chance of using a voucher within a shorter period of time, but has no effect on the probability of the household ever using their voucher, at least on average.

In the models that follow, I summarize the effect of day of issuance using an indicator variable for voucher issuance in the last seven days of the month:

$$Success_{itdp} = \beta End\,of\,month_{id} + \ \gamma X_i' + PHA_p \times Year_t + \varepsilon \tag{2}$$

Again, I use PHA by year fixed effects to hold constant allowable search time policies, and I control for the same set of individual characteristics as in equation 1.

In Figure 4, I show the effect of end of month issuance at every relevant search interval. There are negative, significant effects at 60, 90, and 120 days, and a marginal effect at 150 days, before effects attenuate to zero at longer search windows. Relative to searches that begin in the first three weeks of the month, searches that begin at the end of the month are around 2 percentage points less likely to succeed within 60 days, 1 percentage point less likely to succeed in 90 days, and .75 percentage points less likely to succeed in 120 days. Even at 150 days, we see a close to significant effect of end of month issuance.

These results reflect both the difficulty of using vouchers, and the mechanical relationship between voucher issuance and norms in the private housing market. The fact that there are significant effects even beyond the 30- and 60-day windows where we would expect the effects of issuance timing to be most pronounced emphasizes how difficult it can be to complete all the steps of program participation and start a lease within a certain time period, given that the housing market is structured to start leases on the first of the month. In other words, the date of issuance matters for the ability of voucher holders to use their vouchers and start their leases within certain time frames.

One question is how this effect has evolved over time. On one hand, as housing markets have gotten tighter and the housing search more difficult, there could be a larger effect of losing additional search days. However, it's also possible that in the high search time environment of the past several years, the 60 day threshold has become less salient, so the effects may be minimal. Figure 5 presents coefficients from three different stratified models across different time periods, again using equation 2.

Consistent with expectations, in earlier years (2015-2017), there is a larger penalty for end of month issuance on 60-day success rates. However, when looking at longer search windows, effect sizes are more consistent over time. In 2021 and 2022, there is a small (about half a percentage point) but significant effect even at the 150-day time frame. This suggests that as search times have increased, it's more difficult for households to use their vouchers within 150 days, and an issuance date at the end of the month can therefore make a difference. Past 150 days, there is no significant effect of end of month issuance, though the coefficient for 180-day lease-up is marginally significant in later years, again emphasizing how as searches lengthen, longer search windows become more salient.

A second question concerns heterogeneity in the effects of end-of-month issuance, particularly on the unrestricted measure of success rates. While in the aggregate, we don't see that issuance timing is leading to households losing their vouchers, it's possible that some groups may still struggle to use their voucher if it is issued at the end of the month. In particular, certain groups that face heightened barriers in the housing market, including households of color and households with children, may be most affected by end-of-month issuance.

In Figure 6, I show the coefficients on the end-of-month indicator from stratified models, looking at different individual characteristics. Overall, the effect sizes for 60-day, 90-day and 120-day search windows look largely similar across groups. In some cases, the small deviations we do see in effect sizes may be symptoms of differences in search times across groups. For instance, Black households appear to have a smaller effect for 60-day success; however, Black households also tend to have, by far, the longest search times, so the smaller effect may be due to the fact that very few Black households succeed in using their voucher within 60 days, regardless of when in the month they receive their voucher.

Focusing on the unrestricted measure of success (bottom right quadrant, figure 6), in general, most groups show a precise null effect on the probability of ever using their voucher. The one exception is households over 65, who see a decline of .8 percentage points, which translates to just over a 1% decline. This is somewhat surprising, as in general this group tends to have higher success rates and much shorter search times. However, households over 65 may also have additional challenges in the housing market–for example, difficulty getting to showings or searching for units–that can be more pronounced when they have less time to search.

Finally, the last question I turn to is how to interpret the decline in success rates at shorter search windows. If households are less likely to succeed at certain time intervals, this may indicate that households are experiencing a delay in receiving benefits. However, it's also possible that instead of delaying benefits, households who receive their vouchers at the end the month are not systematically experiencing longer search times; they are simply more likely to miss certain deadlines, such as at 60 day mark, that may be salient cut off points, should a PHA strictly enforce search limits. their leases

To test these hypotheses, I look at the amount of subsidy assistance that a household receives in the 365 days from the date that their voucher is issued. Specifically, I calculate both the number of days that the household is leased up in the program in the year following voucher issuance; I also calculate the daily Housing Assistance Payment (HAP), or the portion of the rent that the PHA pays directly to the landlord, and multiply that amount by the number of days in the year following voucher issuance that a household receives assistance. I use both these measures as the dependent variable in models that follow equation 2.

Table 4 shows the results from these models. Households who receive their voucher at the end of the month, on average, do not receive less assistance over the year following their voucher issuance than those who receive their vouchers at the beginning of the month, measured both in terms of days of assistance and dollar amount of benefits. If anything, we see a slight increase in both days of assistance and dollars of assistance. However, although this increase is statistically significant, it is not economically significant, accounting for just .7 extra days of subsidy, or $20, on average; this translates to less than half a percent of the average annual subsidy that voucher recipients receive.[8] This provides support for the hypothesis that issuance timing is not leading to a delay in benefits, since a portion of households who receive their issuance at the end of the month may begin receiving assistance *earlier* than households who receive their voucher at the beginning of the month, as they reach the next possible lease start date (the 1st of the month) sooner. This means that we can interpret the decrease in 60-, 90-, and 120-day success rates not as a delay in benefits, but as an increase in the probability that a household will fail to use their voucher within the time frame they are given by their PHA, which could result in households losing their voucher altogether if PHAs do not allow households to exceed these time restrictions. This motivates the analysis that follows in section 7, that tests the effects of end-of-month issuance for PHAs with different allowable search time policies.

In Appendix E, I show results from several robustness checks that ensure that these results

[8]Additionally, these results are in general not economically significant, given that unlike entitlement programs where benefit delays are more salient, many of these households have already waiting multiple years to receive assistance.

are not driven by other factors or spurious correlations. First, in table E1, I show that results are robust to omitting households with disabilities and those who receive special purpose vouchers, two groups that may receive additional time or assistance in the search process. Second, I show in table E2 that my results are not sensitive to how I define the end-of-month indicator by using a continuous measure of the day of the month that a voucher is issued, a discrete measure of the week of the month (1-4). Lastly, I show that results are not driven by any sort of seasonal variation, by adding month fixed effects and dropping the month of December (table E3.

6 Consequences for match quality

Given that on average, there is no effect of end of month issuance on the probability of using a voucher at all, I now turn to evaluating the effects of end-of-month issuance on match quality outcomes for participants who are successful in using their vouchers. The expected direction of these outcomes is somewhat ambiguous. If households who receive their issuance at the end of the month are more likely to get an extension, then the extra search time may lead to improved match quality. However, if households are more rushed to lease up in a shorter time period, there may be declines in housing match quality.

I use three different sets of match quality outcomes. First, I look at the probability of a voucher recipient moving ZIP Codes, and the difference between the poverty rates of origin and destination ZIP Codes. A lower rate of moving ZIP Codes could indicate that households are more constrained on search time and are not able to look for units that are further away from where they live. Additionally, households that have less time to search may not be able to make an "opportunity move" to a lower poverty ZIP Codes.

I also look at housing turnover. The length of time that a household remains in a housing unit can be interpreted as a measure of match quality; more frequent moves among households can cause additional search and moving costs both for the households themselves and for PHAs. Specifically, I measure the probability that a household moves housing units within one or two years of entering the program.

Lastly, I look at measures of relative unit size. When households are issued their voucher, they qualify for a certain number of bedrooms based on the size and composition of their household.[9] However, households are allowed to rent homes of different sizes, as long as the home meets the payment standard requirements for the qualified bedroom size. A household is considered "over-housed" if they rent a unit with more bedrooms than they are qualified for. In general, households are not permitted to be "under-housed", or renting a home with fewer bedrooms than they qualify for; however PHAs are allowed to make exceptions. I therefore consider being over-housed as a positive outcome for households who were able to lease larger home, whereas households who are under-housed may have had to sacrifice and rent a smaller home in order to use their voucher.

I estimate the effects of end-of-month issuance on measures of housing match quality using equation 2, with the dependent variable being each binary indicator of match quality described above. In Table 5, I show that end of month issuance is not associated with either expanded neighborhood choice or declines in housing turnover. The coefficient on moving ZIP Codes is significant and negative, indicating that those who receive their vouchers at the end of the month are slightly less likely to move ZIP Codes, though the effect size is quite small, only half a percentage point. This lends support to the idea that some households may be more constrained on search time, and are not able to move in the time period given. Though this does translate to a slight decline in the chance of moving to a *higher* poverty ZIP Code, this is likely just a reflection of households being less likely to move at all.

There is no significant effect on housing unit turnover in the year or two years after admission. Additionally, in the aggregate, there is no effect on the probability of a household being either over- or under-housed. The lack of effects on housing turnover and relative unit size means that while end-of-month issuance may delay program participation for some households, it does not translate into any measurable effect on households using that extra time to find more suitable housing units, and therefore remaining in those homes for longer.

In Appendix F, I show heterogeneity in results on match quality by individual characteristics, again focusing on households that may be more disadvantaged in the search process. In general, there are no strong or consistent trends. I find weak positive effects across some outcomes for Black households; for instance, Black households who receive their issuance at the end of the

[9] PHAs have some flexibility in setting occupancy standards, as long as they are not less generous than federal minimum standards.

month (appendix figure F1) are slightly less likely to move to a higher poverty neighborhood, and are also more likely to lease up into larger housing units (appendix figure F3). Households over 65 are the only category to show negative effects on locational outcomes, and a small uptick in the probability of being under-housed.

7 Variation in effects of issuance timing by PHA policies

To address the question of the effects of issuance timing under different PHA search time policies, I combine the administrative data on success rates with the collected data on search time policies. I use only search events in the years that I have collected plans for a given PHA. Specifically, in this section, I explore if end-of-month issuance can cause declines in success rates at PHAs with stricter search time policies.

The models take the following form:

$$Success_{itdp} = \beta End\,of\,month_d \times Policy\,Feature_{pt} + \ \gamma X_i' + PHA_p + Year_t + \varepsilon \tag{3}$$

I use a fully interacted model that interacts an indicator for end-of-month issuance with a mutually exclusive set of PHA policies, to preserve the PHA fixed effects.[10] Here, I primarily focus on the effect of end-of-month issuance on the probability of a household ever using their voucher, under the hypothesis that a stricter search policy could lead to a decline in unrestricted success rates, given the shorter period of time a household would have to use their voucher before it expires. Again, I control for the same set of individual characteristics, and I use year fixed effects to control for any aggregate trends over time, as most PHAs are observed in only one year between 2015 and 2020.

7.1 Effects on successful voucher use by PHA search policies

First, I test the effect of end-of-month issuance by different initial and effective voucher terms (table 6). The difference between the two terms is that the effective search term adds on the number of days that a household would gain from an automatic extension. For instance, if a PHA

[10] PHA policies may be endogenous to success rates; for this reason, I compare issuances at different times of the month within PHAs.

has an initial search term of 60 days and an automatic extension of 60 days, their effective initial search term is 120 days. We may expect here that at PHAs that offer only 60 days, end-of-month issuance may lead to a decline in unrestricted success rates, given the decline in 60-day success rates documented in section 5.

None of the coefficients are statistically significant. This may be due to the fact that nearly all PHAs allow for extensions, including additional discretionary extensions on top of any automatic extensions that they may offer. The result of this is that the term of the voucher becomes a less salient deadline, so despite the fact that households may have fewer days of effective search time to lease-up within their 60 day initial search term, they have the ability to easily exceed their initial term and not risk losing their voucher.

To account for this, I then test the effect of end-of-month issuance by type of extension policy. I group PHAs into three mutually exclusive categories: PHAs that offer no extensions, PHAs that offer only discretionary extensions, and PHAs that offer both automatic and discretionary extensions.[11] Table 7 shows these interaction terms, both for all PHAs and for PHAs that have a 60 day initial search term. There are strong effects of end-of-month issuances at the most stringent set of PHAs: those that offer 60-day initial search terms, and do not offer any type of extensions. Households who receive their vouchers in the last seven days of the month are 5.7 percentage points (7%) less likely to ever use their vouchers at this set of PHAs. This is a large and meaningful decline, showing that at more lenient PHAs, the ability to receive an extension mitigates the effects of any loss in effective search time that can accompany end-of-month voucher issuance. Put another way, at PHAs with the strictest search time policies, issuance timing can affect the ability of households to use their vouchers, while at PHAs that allow for either discretionary or both automatic and discretionary extensions, there is no effect of end-of-month issuance on the probability of the household ever using their voucher, even at PHAs that give the shortest possible 60-day initial term.

Again, these results raise the issue of targeting within the program; or who is able to actually use their vouchers and receive assistance. In Table 8, I show results from stratified models for Black households and households with children, two groups that may face higher barriers in the housing market (Faber and Mercier, 2022; Briggs et al., 2010). Due to the smaller sample size of

[11] All PHAs that allow automatic extensions also allow for discretionary extensions.

the allowable search time sample and the relative few PHAs with more stringent policies (23 PHAs have both a 60 day search time and a no extension policy), I focus on these two characteristics where the sample is large enough to yield reliable estimates. I include only PHAs with a 60-day initial term, in order to capture the set of more stringent PHAs.

While insignificant, coefficients on the interaction term between end-of-month issuance and the indicator for a PHA not allowing for any type of extension are quite similar across Black and white households. However, focusing on the last two columns, effects are strongest for households with children. This is consistent with larger households, and households with kids, facing additional hurdles in the housing search and voucher take-up process, which can be exacerbated by a reduced amount of time to complete all the steps in the process. These results indicate that additional flexibility in time constraints can make the program more accessible for families with children.

7.2 Effects on match quality by PHA search policies

One reason for the lack of strong effects on match quality in the aggregate is that effects on match quality could vary by PHA search time policies. For instance, PHAs with shorter initial search times or less flexible extension policies may see negative effects on match quality, as households contend with a shorter effective search period. For PHAs with longer initial search times or more flexible extension policies, effects are more ambiguous. There is also reason to expect no effect, as most households are still able to complete their search in the given time frame, even though they may have fewer effective days to search. On the other hand, households who know they have a longer period of time to search may aim to start a lease a month later, given their later issuance date, and therefore may look at a broader set of housing options or may be more likely to hold out for a housing unit that is a better match.

In order to understand the effects on housing match quality, I first look at the average effects of end-of-month issuance on observed search time for households are PHAs with different search time policies, since search time is the mechanism through which we would except issuance timing to have an effect on different measures of housing match quality. As in the descriptive statistics, search time is measured as the elapsed time between the issuance of a voucher and the household leasing up into the program, for households who are successful in their search. In Table 9, I use equation 3 to show the average effects of end-of-month issuance on observed search time, by both initial search terms and extension policies.

Focusing first on initial search time, we see that on average, there are no effects on search time across any of the three most common initial search terms. Again, the effects of end-of-month issuance are most pronounced for more stringent PHAs. PHAs with 60-day search times that do not offer extensions see the large declines in search time. This set of PHAs with the strictest search time policies see average search times for vouchers that were issued at the end of the month that are over around 8 days (10%) shorter than those issued in the first three weeks. This set of PHAs would likely be most susceptible to seeing declines in housing match quality caused by end-of-month issuance, given that it is translating into significant declines in search time.

In Table 10, I again use equation 3 to test the effects of end-of-month issuance on measures of match quality by PHA initial search term. Overall, there are no strong or consistent effects on match quality, across either of the three groups. For PHAs with a 60-day initial term, there are no effects on neighborhood outcomes, unit turnover, or relative unit size. At PHAs with 90-day initial terms, there is a 3.1 percentage point (10%) decline in the probability of a successful voucher recipient moving ZIP Codes with their voucher. This translates into these households also being less likely to move to a lower poverty neighborhood. At PHAs with a 120-day initial term, we see a decline in the probability of a household moving housing units within 2 years of entering the program. Combined, these results mean that households at PHAs with 90-day initial terms may see reductions in their ability to move to certain neighborhoods when they receive a voucher at the end of the month, while those at PHAs with longer initial search times see slight increases in match quality.

In Table 11, I test the effect of end-of-month issuance by extension policy, again using mutually exclusive categories of no extensions, discretionary extensions only, or both discretionary and automatic extensions. One striking results is that households at PHAs that do not offer extensions are more 1.8 percentage points likely to be under-housed, again reflecting a more constrained search process. This translates to an economically meaningful 18% increase of the base rate; being under-housed is uncommon, as in general it requires permission from the housing authority. This effect is in contrast to PHAs that allow discretionary extensions, where we see a decline in the probability of being under-housed. This is again reflecting the fact that at stricter PHAs, end-of-month issuance can have more substantial effects on how long a voucher recipients searches for and therefore their ability to find a unit that meets their needs. When these timelines are more condensed, households are more likely to end up in over-crowded housing situations, which is an important result given that reducing the prevalence of over-crowded housing situations is a core goal of the voucher program.

8 Discussion

In sum, I find that while, on average, issuance timing does not influence the ability of households to ever use their vouchers, there are significant declines in the ability to use a voucher within certain time periods. While many PHAs have built in the flexibility in their search time policies to mitigate any lasting negative effects of end-of-month issuance, at PHAs that offer limited flexibility in voucher term, end-of-month issuance can cause a substantial decline in the ability of households to use their vouchers. There is also suggestive evidence that end-of-month issuance combined with a lack of extensions can limit search time, leading to voucher recipients leasing up into smaller housing units when they are successful.

This work has direct policy implications. First, descriptive findings emphasize the growing difficulty that voucher holders experience in using their vouchers, and high search costs that even successful households must endure. While the relationship between allowable search time and success rates is difficult to disentangle, this paper provides the first evidence that allowable search time policies can have meaningful impacts on certain elements of the housing voucher take-up process and potentially on match quality for successful recipients as well.

An encouraging finding is that many PHAs are already giving participants the flexibility in search times to enable successful voucher use, which overall works to mitigate any potential long-term negative effects of receiving a voucher at the end of the month. The findings in this paper emphasize that extensions are key in enabling households to navigate both the program take-up process and the housing market search process in order to use their vouchers. When extensions are not permitted, PHAs need to be cognizant of issuance timing, and adopt either issuance strategies or more flexible search time policies that can alleviate the effects of end-of-month issuance.

The second policy implication concerns the overall difficulty that voucher recipients have in using their vouchers. More than two in five voucher recipients must return their vouchers to the housing authority in recent years, even despite the fact that many housing authorities already offer flexible search time policies. Strategies such as passing and enforcing Source of Income discrimination laws, reforming inspection practices, and potentially allowing households to move into housing units before inspections are completed when housing markets are tight are important to boost overall success rates and ensure that other factors such as issuance timing don't interfere with the ability of households to use their vouchers.

The collected data on allowable search time policies also opens up important avenues for future research. First, although at this point the administrative data on success rates does not allow for analysis past 2022, through the collected plans I observe that many PHAs have extended their initial search terms, or imposed additional flexibility in the form of automatic extensions in 2023 or 2024. In the future, these policy changes will provide the opportunity to more directly evaluate these endogenous policies by looking at changes over time within a PHA. Second, given recent significant declines in success rates, if HUD chooses to raise the minimum 60-day initial search term, the collected policies will provide critical information on where this potential increase is binding, which will be essential for understanding how extended search time could increase success rates.

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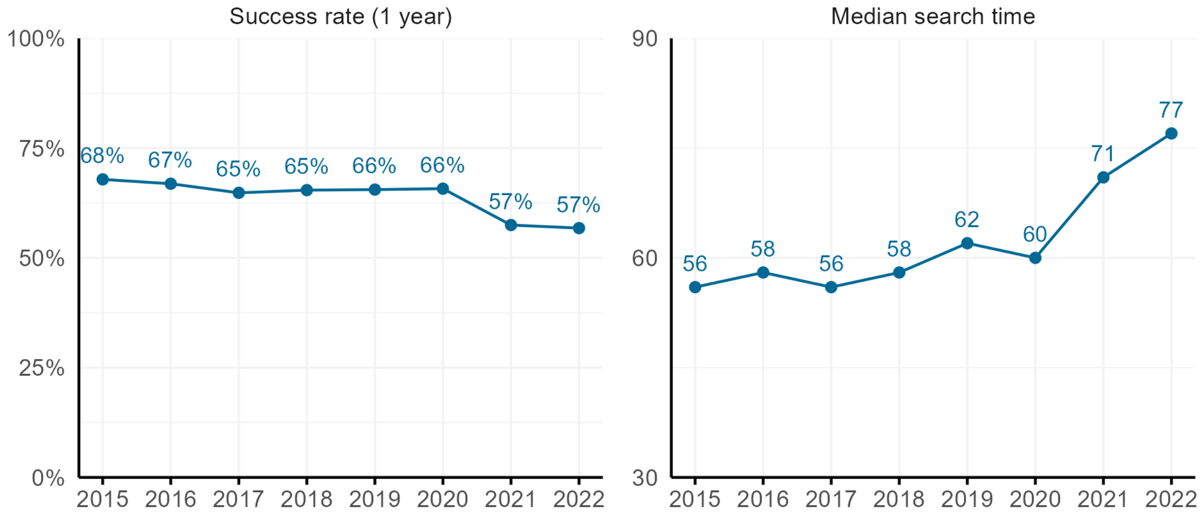
Tables and figures

Table 1: Sample summary statistics

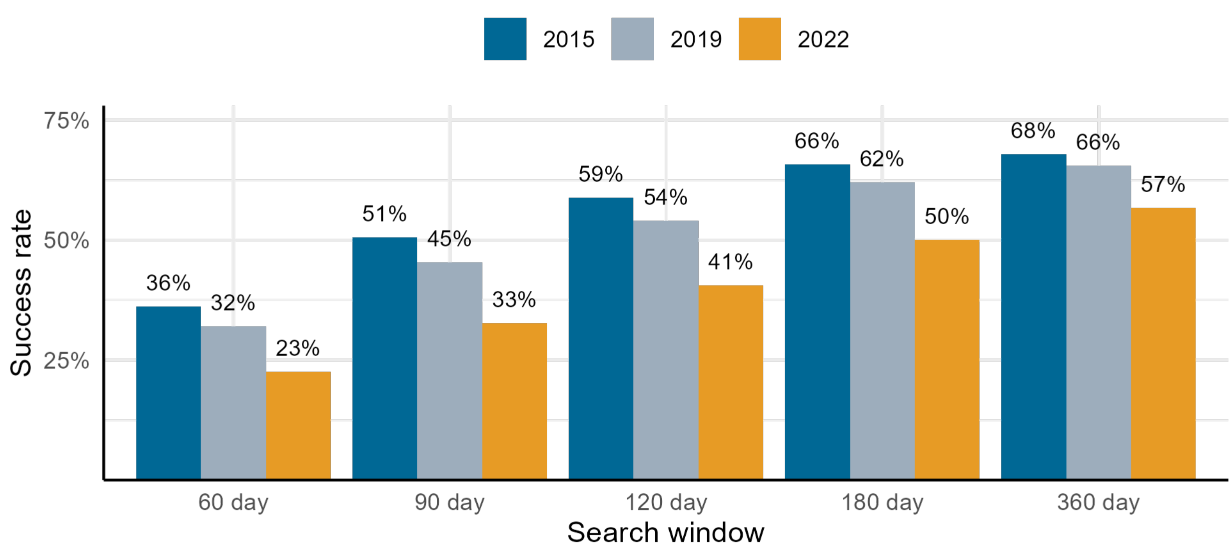
	Observations	Share of observations	Success rate (1-year)	Median search time
Full administrative data sample				
Black head of household	448,797	0.459	0.647	75
Hispanic head of household	139,128	0.142	0.651	63
White head of household	341,000	0.349	0.630	49
Other race head of household	48,193	0.049	0.614	59
Female head of household	708,760	0.725	0.644	64
Presence of kids	478,653	0.490	0.651	67
Age 65+	100,885	0.103	0.678	46
Disability indicator	308,375	0.316	0.646	59
Homeless indicator	120,012	0.123	0.645	71
All observations	1,112,239		0.635	62
Allowable search time sample				
Black head of household	43,152	0.546	0.667	87
Hispanic head of household	14,650	0.185	0.626	75
White head of household	17,145	0.217	0.619	63
Other race head of household	4,073	0.052	0.564	70
Female head of household	57,257	0.725	0.664	79
Presence of kids	43,198	0.547	0.680	79
Age 65+	6603	0.084	0.653	57
Homeless indicator	12,971	0.164	0.575	101
All observations	91,529		0.646	78

Figure 1: Success rates and search times, 2015-2022



Note: The panel on the left shows the 1-year success rates, or the share of voucher recipients who are successful in using their voucher within one year of issuance. The panel on the right shows the median search time, or the median elapsed time between voucher issuance and successful lease-up.

Figure 2: Success rates at different search windows, 2015, 2019, and 2022



Note: The success rate at each search window is defined as the share of households who are successful in using their voucher within the search window given.

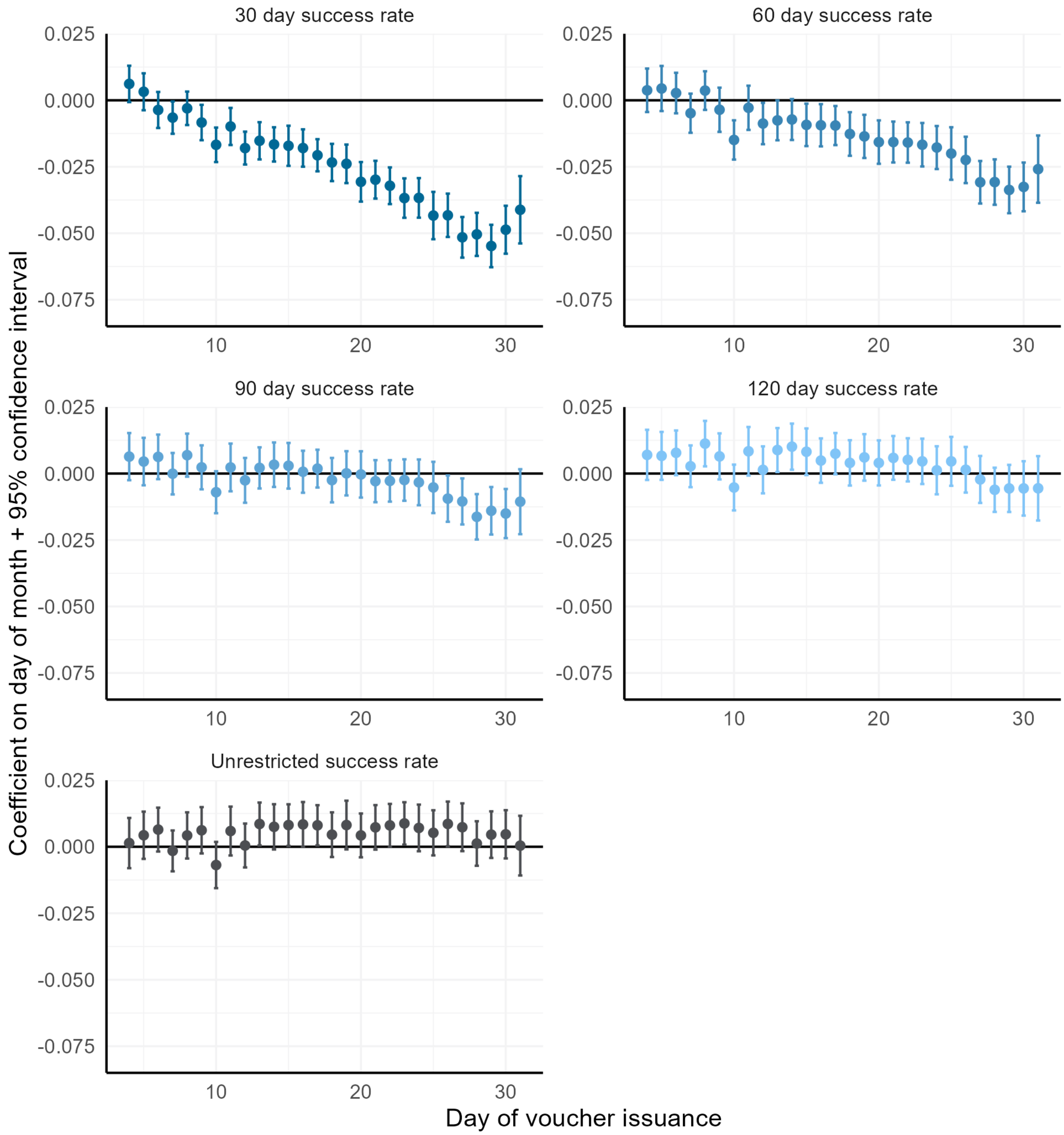
Table 2: PHA search time policy details

Initial voucher term	Number of PHAs	Share of PHAs that do not offer extensions	Share of PHAs that offer discretionary extensions only	Share of PHAs that offer automatic and discretionary extensions	Share of PHAs with a maximum term limit	Share that suspend vouchers during unit inspection
Varies	14	0.077	0.692	0.214	0.571	1.000
60 days	397	0.055	0.650	0.297	0.474	0.919
90 days	46	0.022	0.696	0.283	0.283	0.929
120 days	50	0.140	0.780	0.080	0.260	0.958

Table 3: Success rates and search time by PHA initial search term policy

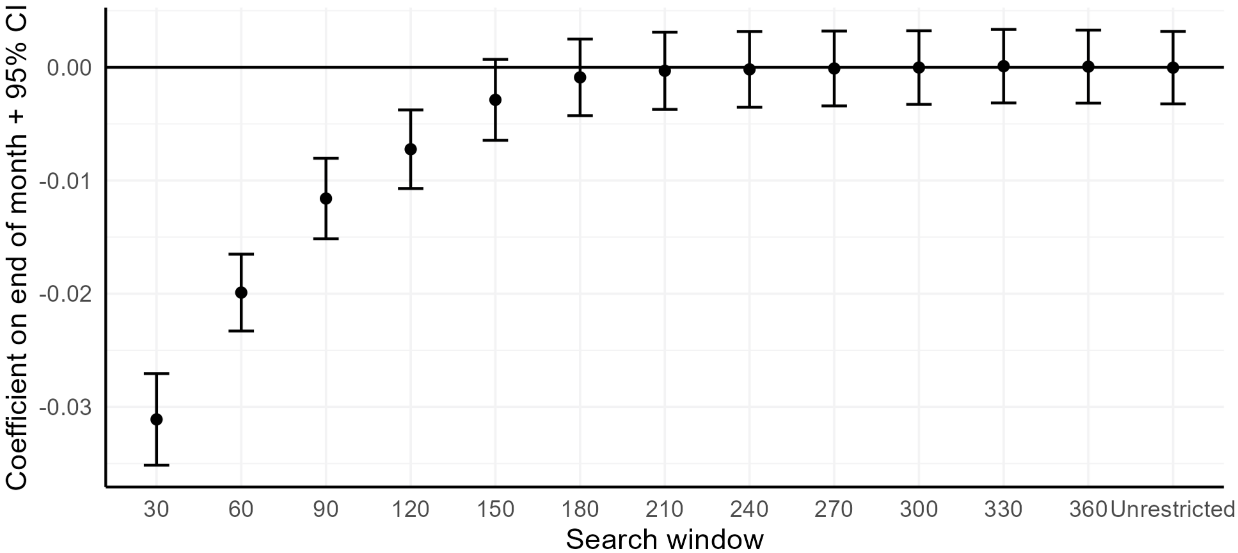
Initial voucher term	Number of search events	Success rate (1 year)	Median search time	Share of searches at least 15 days longer than intitial voucher term	Share of searches at lease 15 days longer than effective initial voucher term
60	77380	0.642	80	0.535	0.487
90	8867	0.661	76	0.288	0.257
120	5001	0.676	71	0.205	0.202

Figure 3: Effect of day of issuance on 30-day 60-day, 90-day, 120-day, and unrestricted search window measures of success



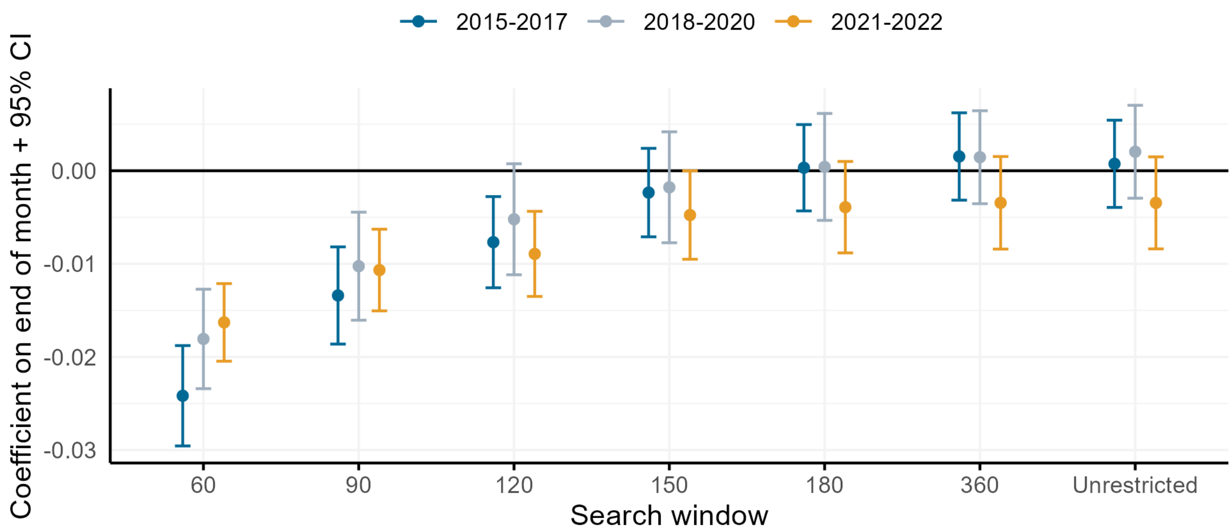
Note: The dependent variable in each frame in an indicator for the household successfully using their voucher within the given search window. Each point represents the coefficient on an indicator for the voucher being issued on that day of the month. Models include individual controls (race/ethnicity, age, household composition, disability status, homeless status, citizenship status) and PHA by year fixed effect to hold allowable search time policies constant. Standard errors are clustered by PHA.

Figure 4: Effect of end-of-month issuance on success at different search windows



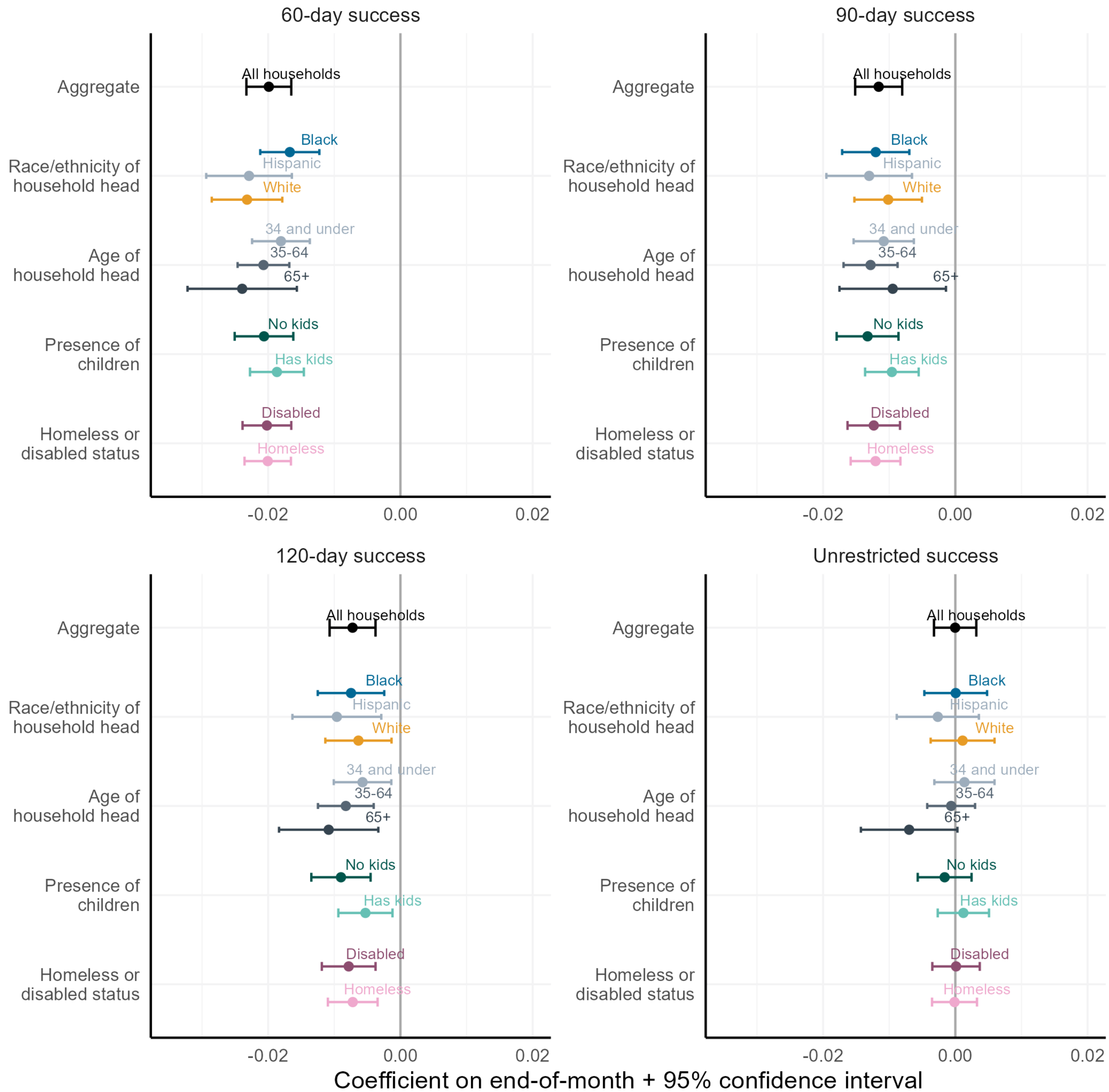
Note: Each point represents the coefficient on the end-of-month indicator variable; the dependent variable is an indicator for the household leasing up within a given search window. Models include individual controls (race/ethnicity, age, household composition, disability status, homeless status, citizenship status) and PHA by year fixed effect to hold allowable search time policies constant. Standard errors are clustered by PHA.

Figure 5: Effect of end-of-month issuance on success at different search windows, over time



Note: Each point represents the coefficient on the end-of-month indicator variable; the dependent variable is an indicator for the household leasing up within a given search window. Models are stratified by time period. Models include individual controls (race/ethnicity, age, household composition, disability status, homeless status, citizenship status) and PHA by year fixed effect to hold allowable search time policies constant. Standard errors are clustered by PHA.

Figure 6: Heterogeneity in effect of end-of-month issuance on success at different search windows



Note: Each point represents the coefficient on the end-of-month indicator variable; the dependent variable is an indicator for the household leasing up within a given search window. Each point represents the coefficient from a stratified model. Models include all other individual controls (race/ethnicity, age, household composition, disability status, homeless status, citizenship status) and PHA by year fixed effect to hold allowable search time policies constant. Standard errors are clustered by PHA.

Table 4: Average days of assistance and dollars of assistance received in year following voucher issuance

	Days of assistance		Dollars of assistance	
	(1)	(2)	(3)	(4)
End of month indicator	0.696*		20.915*	
	(0.281)		(10.641)	
Week 2 of month		-0.189		-24.937*
		(0.240)		(12.413)
Week 3 of month		0.180		-9.489
		(0.253)		(12.001)
Week 4 of month		0.787*		4.331
		(0.308)		(11.722)
Observations	666,163	666,163	661,023	661,023
Adjusted R^2	0.210	0.210	0.635	0.635
PHA x Year fixed effects	X	X	X	X
Individual controls	X	X	X	X
Controls for household size and income			X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Models include only households who were successful in their search. The dependent variable in the first two columns is the number of days that the household received assistance in the 365 days that followed the issuance of a search voucher. The dependent variable in the second set of columns is the dollar amount of this assistance, calculating by taking the monthly subsidy payment, calculating the daily amount, and multiplying by the number of days that the household was assisted. The second two columns control for household size and income, which affect the dollar amount of the subsidy payment. Households who are homeless at the time of voucher issuance are not included in neighborhood models, as it's unclear how to interpret their origin ZIP Code. All models include all other individual controls (race/ethnicity, age, household composition, disability status, homeless status, citizenship status) and PHA by year fixed effect to hold allowable search time policies constant. Standard errors are clustered by PHA.

Table 5: Effect of end-of-month issuance on measures of housing match quality

	Move ZIP Codes	Move to lower poverty ZIP Code	Move to a higher poverty ZIP COde	Move within 1 year	Move within 2 years	Over-housed	Under-housed
End of month	-0.004*	-0.001	-0.003*	0.000	-0.001	0.001	0.000
	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)	(0.001)
Observations	539,425	522,788	522,788	676,888	676,888	594,004	594,004
Adjusted R^2	0.192	0.096	0.058	0.014	0.030	0.118	0.041
PHA x Year fixed effects	X	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in each column is the binary indicator for match quality given in the column header. Those who are homeless at the time of voucher issuance are excluded from models with locational outcomes. Individual controls include race/ethnicity, age, household composition, disability status, homeless status (where applicable) and citizenship status.

Table 6: Effect of end-of-month issuance on unrestricted measure of success, by PHA initial and effective search terms

	Effect on successful voucher use (unrestricted)	
	Initial search term	Effective search term
End of month x 60 day initial term	0.001	
	(0.005)	
End of month x 90 day initial term	0.022	
	(0.020)	
End of month x 120 day initial term	-0.034	
	(0.026)	
End of month x 60 day effective term		0.000
		(0.007)
End of month x 90 day effective term		0.000
		(0.012)
End of month x 120 day effective term		0.000
		(0.016)
Observations	84,215	84,215
Adjusted R^2	0.055	0.055
PHA fixed effects	X	X
Year fixed effects	X	X
Individual controls	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in both columns is an indicator for the voucher recipient ever successfully using their voucher. Disabled households are excluded from the model, as they are frequently subject to different allowable search time policies. Individual controls include race/ethnicity, age, household composition, homeless status (where applicable) and citizenship status.

Table 7: Effect of end-of-month issuance on unrestricted measure of success, by PHA extension policies

	All PHAs	PHAs with 60 day initial terms
	(1)	(2)
End of month x No extensions	-0.032	-0.057*
	(0.022)	(0.028)
End of month x Discretionary extensions only	0.000	0.001
	(0.007)	(0.007)
End of month x Automatic and discretionary extensions	0.004	-0.002
	(0.008)	(0.010)
Observations	84,215	73,183
Adjusted R^2	0.055	0.057
PHA fixed effects	X	X
Year fixed effects	X	X
Individual controls	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in both columns is an indicator for the voucher recipient ever successfully using their voucher. Disabled households are excluded from the model, as they are frequently subject to different allowable search time policies. Individual controls include race/ethnicity, age, household composition, homeless status (where applicable) and citizenship status.

Table 8: Effect of end-of-month issuance on unrestricted measure of success for PHAs with 60-day initial search terms, stratified by race and presence of children

	Race of household head		Presence of children	
	Black head of household	White head of household	Has kids	No kids
End of month x No extensions	-0.050	-0.052	-0.056***	-0.023
	(0.037)	(0.040)	(0.016)	(0.049)
End of month x Discretionary extensions only	0.001	-0.014	0.006	-0.003
	(0.008)	(0.011)	(0.010)	(0.007)
End of month x Automatic and discretionary extensions	-0.007	0.000	-0.001	-0.004
	(0.012)	(0.017)	(0.009)	(0.014)
Observations	38,114	16,439	39,695	33,488
Adjusted R^2	0.054	0.060	0.066	0.051
PHA fixed effects	X	X	X	X
Year fixed effects	X	X	X	X
Individual controls	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in all columns is an indicator for the voucher recipient ever successfully using their voucher. Disabled households are excluded from the model, as they are frequently subject to different allowable search time policies. Individual controls include race/ethnicity, age, household composition, homeless status (where applicable) and citizenship status.

Table 9: Effect of end-of-month issuance on observed search time, by PHA initial search term and extension policies

	All PHAs		PHAs with 60 day initial term
	(1)	(2)	(3)
End of month x 60 day initial term	-0.309		
	(1.374)		
End of month x 90 day initial term	-0.739		
	(3.330)		
End of month x 120 day initial term	1.462		
	(3.090)		
End of month x No extensions		-3.601	-7.935*
		(3.391)	(3.316)
End of month x Discretionary extensions only		-0.614	-0.402
		(1.431)	(1.492)
End of month x Automatic and discretionary extensions		2.483	1.699
		(2.798)	(3.091)
Observations	40,759	40,759	35,128
Adjusted R^2	0.092	0.093	0.097
PHA fixed effects	X	X	X
Year fixed effects	X	X	X
Individual controls	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in all columns is the elapsed time, in days, between voucher issuance and successful lease-up, for households who are successful. Disabled households are excluded from the model, as they are frequently subject to different allowable search time policies. Individual controls include race/ethnicity, age, household composition, homeless status (where applicable) and citizenship status.

Table 10: Effect of end-of-month issuance on measures of housing match quality, by initial search term

	Move ZIP Codes	Move to lower poverty ZIP Code	Move to higher poverty ZIP Code	Move within 1 year	Move within 2 years	Over-housed	Under-housed
End of month x 60 day initial term	-0.003	0.000	-0.002	-0.001	-0.002	0.007	-0.007
	(0.006)	(0.007)	(0.008)	(0.002)	(0.005)	(0.006)	(0.004)
End of month x 90 day initial term	-0.031*	-0.020+	-0.014	0.001	-0.001	0.039	-0.004
	(0.015)	(0.012)	(0.014)	(0.007)	(0.009)	(0.028)	(0.007)
End of month x 120 day initial term	-0.020	-0.007	-0.013	-0.012	-0.031*	-0.023	-0.005
	(0.026)	(0.025)	(0.015)	(0.009)	(0.014)	(0.016)	(0.015)
Observations	34,838	34,251	34,251	40,759	40,759	40,536	40,536
Adjusted R^2	0.128	0.063	0.033	0.014	0.032	0.085	0.032
PHA fixed effects	X	X	X	X	X	X	X
Year fixed effects	X	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in each column is the binary indicator for match quality given in the column header. Those who are homeless at the time of voucher issuance are excluded from models with locational outcomes. Disabled households are excluded from the model, as they are frequently subject to different allowable search time policies. Individual controls include race/ethnicity, age, household composition, homeless status (where applicable) and citizenship status.

Table 11: Effect of end-of-month issuance on measures of housing match quality, by type of extension policy

	Move ZIP Codes	Move to lower poverty ZIP Code	Move to higher poverty ZIP Code	Move within 1 year	Move within 2 years	Over-housed	Under-housed
End of month x No extensions	-0.015	-0.023	0.003	-0.001	0.007	-0.023	0.018*
	(0.026)	(0.027)	(0.017)	(0.008)	(0.024)	(0.028)	(0.008)
End of month x Discretionary extensions only	-0.003	0.001	-0.004	0.000	-0.004	0.007	-0.010**
	(0.006)	(0.007)	(0.007)	(0.002)	(0.005)	(0.007)	(0.003)
End of month x Automatic and discretionary extensions	-0.023	-0.013	-0.009	-0.010+	-0.008	0.012	0.005
	(0.021)	(0.017)	(0.024)	(0.005)	(0.007)	(0.010)	(0.012)
Observations	34,838	34,251	34,251	40,759	40,759	40,536	40,536
Adjusted R^2	0.128	0.063	0.033	0.014	0.032	0.083	0.031
PHA fixed effects	X	X	X	X	X	X	X
Year fixed effects	X	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in each column is the binary indicator for match quality given in the column header. Those who are homeless at the time of voucher issuance are excluded from models with locational outcomes. Disabled households are excluded from the model, as they are frequently subject to different allowable search time policies. Individual controls include race/ethnicity, age, household composition, homeless status (where applicable) and citizenship status.

Appendix

Appendix A: Administrative data on voucher use: data quality standards

This paper uses administrative data from HUD to derive outcomes on success rates and search time. The process to clean these data was extensive; it is described in detail in Ellen et al. (2023). In order to ensure accuracy, I employ data quality standards to ensure that only PHAs with accurate and complete issuance data are included in the sample. In general, data quality standards ensure that I exclude PHAs that do not accurately report their issuance data; these PHAs are likely to have success rates that are biased upwards, as they could be under-reporting failed searches.

There are 1,213 PHAs that meet data quality standards across the seven year panel period from 2015-2022. These PHAs collectively serve around 71% of voucher holders. I then further restrict the sample to PHAs that are serve voucher holders that live predominately in one county, in order to get an accurate measure of housing market conditions; 1,810 PHAs meet this criteria, or 73% of the 2,202 PHAs that served tenant-based voucher holders during this time period. Initial Moving-To-Work (MTW) PHAs are also excluded; these PHAs have different reporting requirements, and do not report origin ZIP Code, which is used to derive some of the key outcomes.

Combining these restrictions results in a final sample of 950 PHAs, which collectively cover 47% of voucher holders (using the population of voucher holders in 2019). Table A1 shows that searches that originate at these PHAs look very similar to all PHAs, and all PHAs that meet data quality standards.

Table A1: Balance test for inclusion in sample

Variable	All PHAs	PHAs that meet data quality standards	PHAs predominately covering one county	PHAs in final sample
Black head of household	0.426	0.444	0.445	0.452
Hispanic head of household	0.145	0.140	0.145	0.146
White head of household	0.384	0.371	0.361	0.355
Female head of household	0.726	0.724	0.721	0.723
Presence of children	0.477	0.476	0.476	0.483
Age 65+	0.113	0.110	0.114	0.111
Disabled head of household	0.320	0.322	0.319	0.319
Noncitizen	0.026	0.025	0.027	0.026
Homeless indicator	0.105	0.106	0.108	0.114
Observations	2,207,503	1,712,372	1,608,844	1,036,534

Appendix B: Collecting data on allowable search times

While PHAs are mandated to give voucher recipients 60 days to search when they receive a voucher, there is no maximum amount of time, and PHAs have broad discretion to set initial search time and extension policies. There is, however, no centralized source of information about these policies. From HUD's Housing Choice Voucher guidebook:

> PHAs must give all voucher holders a minimum of 60 days search time. There is, however, no limit on how much longer a PHA may allow a family to search for housing. There are differing opinions on what length of search time is effective in promoting family success in leasing.

HUD also notes that PHAs that use longer search times should maintain data on searches so that they can monitor if extended search times could improve success rates, and to have knowledge of their potential financial liabilities. To my knowledge, this is the first paper that provides descriptive evidence on search time policies.

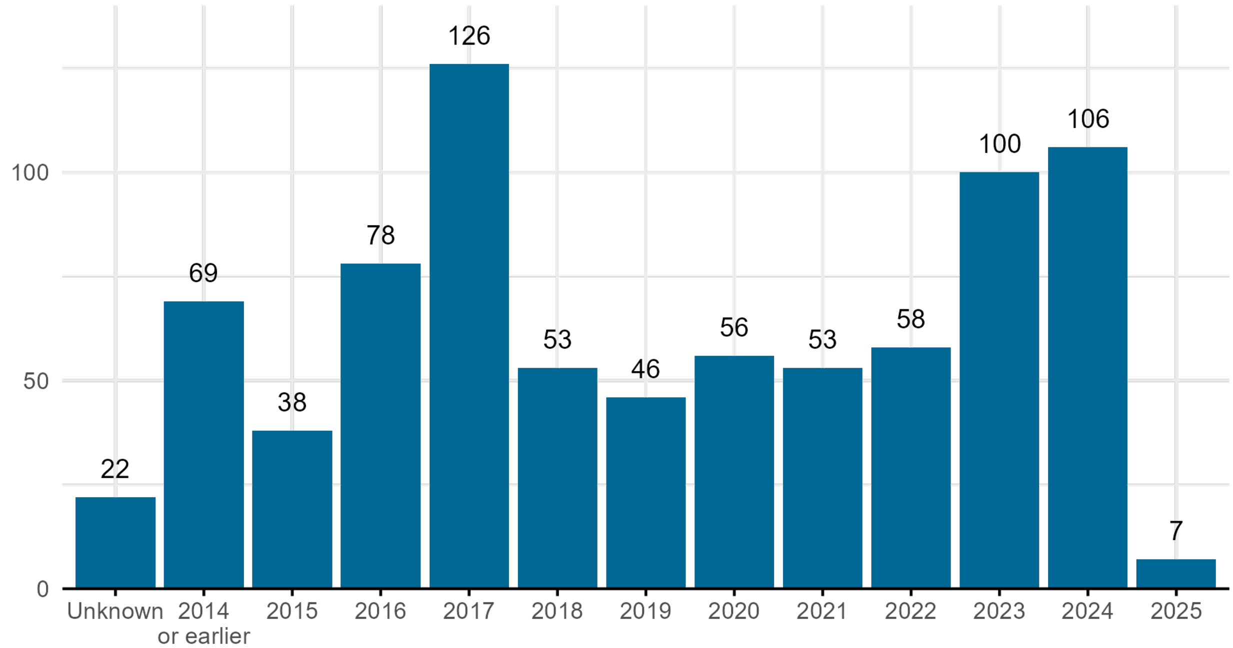
Each year, PHAs submit administrative plans, where they document a comprehensive set of policies that guide the day-to-day operations of the programs. PHAs are required to document their initial search time, or how many days the voucher is issued for; their policies around extensions, which can include automatic and discretionary extensions; their policies on voucher suspension; and their policies around expiration, or what happens when a household fails to use their voucher in the allotted time given.

For this paper, I collect data on these policies by collecting administrative plans from several sources. First, Zhang and Johnson (2023) provided over 1,200 administrative plans and other documents from PHAs, collected between 2016 and 2020. To my knowledge, this is one of the only sources of historical plans, as most PHAs post only their more recent administrative plans on their websites. Of the documents provided, 402 PHAs are in my administrative data sample, meaning they meet data quality standards for the time period. For these PHAs, I can observe their search time policies, and see how these policies are reflected for both failed and successful searches.

In addition to the historical plans, I collected 410 administrative plans directly from PHA websites, or in some cases, through email correspondence with PHAs. Since the data collection process took place between March and September 2024, most plans that were collected from websites cover later time periods, so are not able to be linked with the administrative data on success rates; however, these plans are important for understanding what sort of allowable search time policies are most common today, and how these policies have shifted over time. A select few PHAs posted multiple years of plans on their websites, where I am able to create a more complete history of the PHAs search time policies; however, this was relatively rare.

In Figure B1 below, I show the distribution of the effective year of the analyzed plans. I focused on collecting plans for PHAs that I could observe in multiple time periods. This usually entailed starting with a historical plan from Zhang and Johnson (2023), and then looking to the PHAs website to find a more recent update. As a result of this, I observe most PHAs at multiple points in time. I have two or more plans for just over 60% of the 433 unique PHAs in my sample. However, in general, I only observe a PHA once during my administrative data period, since typically, at least one of each PHAs plans are from the years 2023, 2024, or 2025.

Figure B1: Effective year of plans analyzed



In the analysis that follows, I exclude plans with an unknown year. In the paper, I focus on the set of PHAs that have either 60, 90, or 120 day initial search term, which constitutes the vast majority of PHAs in my sample. However, there is a small set of PHAs that give a variable initial search term.[12] A few PHAs give longer than 120 days as an initial term, and only one PHA appears to be out of compliance with HUD regulations, offering only a 30 day initial term.

Table B1: Number of PHAs, by initial voucher search term

	Number of plans	Share of plans
Varies	20	0.02
30 days	1	0.00
60 days	591	0.73
90 days	79	0.10
120 days	114	0.14
180 days	6	0.01
210 days	1	0.00

[12] Most commonly, these plans will have language that indicates vouchers can be issued for a term of either 60 days or 120 days. In some cases, the initial voucher term is to be determined based on market conditions, such as vacancy rates.

However, the initial search term alone does not paint the complete picture of allowable search time policies, since nearly all PHAs allow for either discretionary or automatic extensions. An automatic extension is an extension that households must request from the PHA, but the request will not be denied; in other words, automatic extensions do not require the household to meet any set of conditions. Discretionary extensions, on the other hand, are extensions that can be granted by the PHA, but the household may need to meet one or more conditions. Common reasons for extensions include the household showing evidence that they have been searching, an employment or family emergency, or tight rental market conditions. All PHAs must allow for additional search time as a reasonable accommodation for households with disabilities. PHAs will also often provide extra search time at issuance to households receiving certain types of special purpose vouchers. Across all the plans analyzed, 51 (6%) indicate that the PHA does not allow for any extensions.

I combine the information on initial terms and automatic extensions to create the effective initial search term, which takes the initial term and adds the number of days that would be allotted from any automatic extensions. Although it's not clear if participants are told that these extensions are indeed automatic, the effective term may capture the conditions surrounding search time more accurately if either participants are aware of them, or a high share of participants take up these extensions.

Table B2: Number of PHAs, by effective initial search term

	Number of plans	Share of plans
30	1	0.00
60	418	0.53
90	199	0.25
120	146	0.18
150	14	0.02
180	12	0.02
210	1	0.00
240	1	0.00

Lastly, I look over time to see how allowable search time policies have evolved over time. Figure B2 shows that between 2014 and 2024, the share of PHAs that offer the federal minimum 60-day term has dropped considerably. From 2014–2016, close to 85% of PHAs offered a 60-day initial term; in 2024, only 56% of PHAs did. Effective terms have gotten longer as well, which is a reflection both of longer initial search times, and more PHAs choosing to enact automatic or discretionary extensions. Figure B3 shows that the share of collected plans that indicate a PHA offers automatic extensions rose between 2017 and 2022; while this share did level off in 2023 and 2024, it remains elevated from the earliest years.

Figure B2: Share of PHAs with a 60-day initial or effective search term, over time

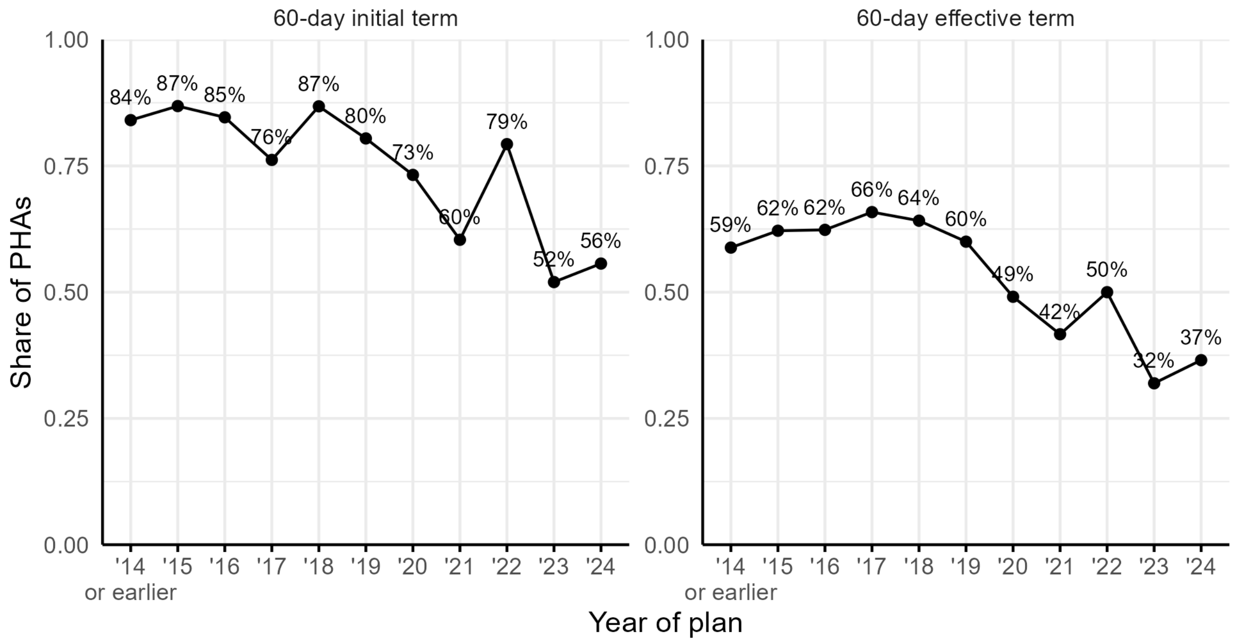
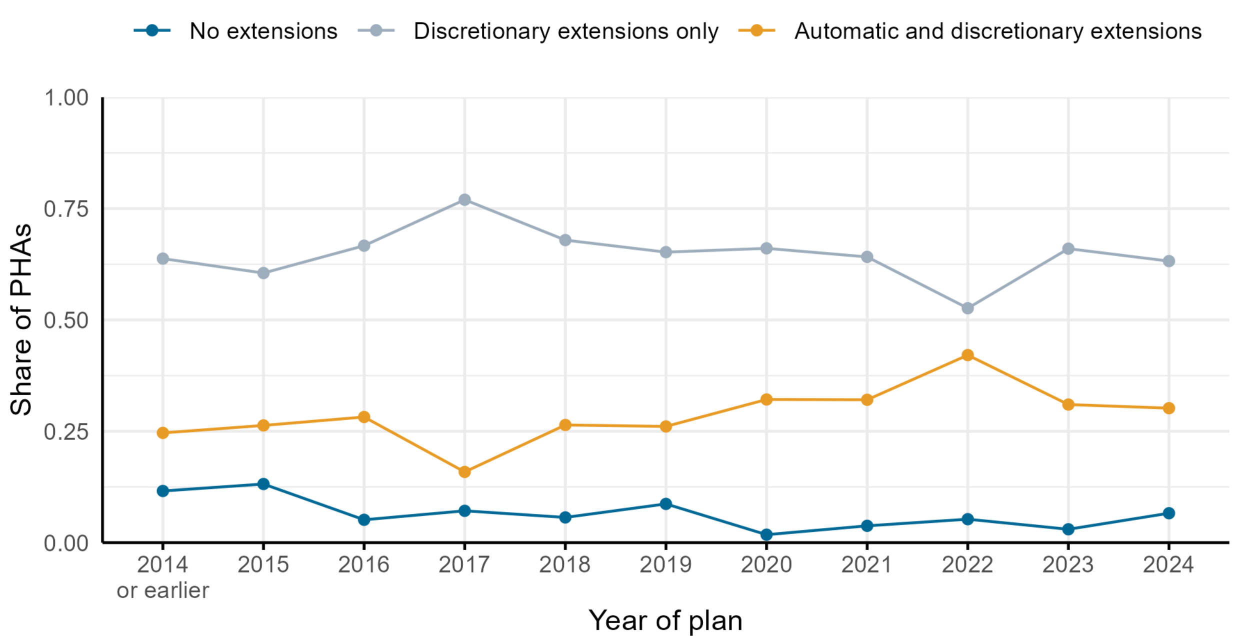
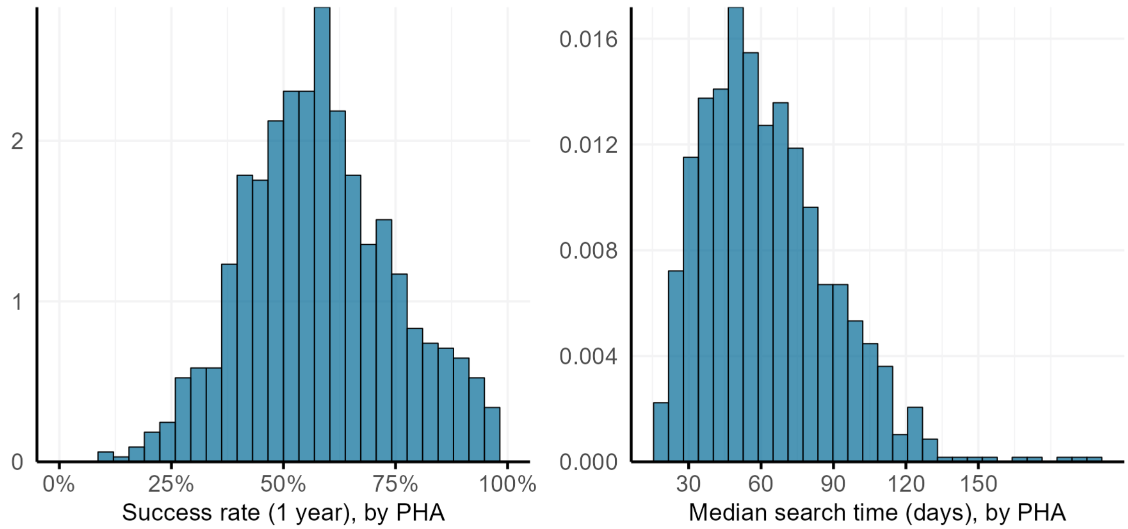


Figure B3: Share of PHAs with each type of extension policy, over time



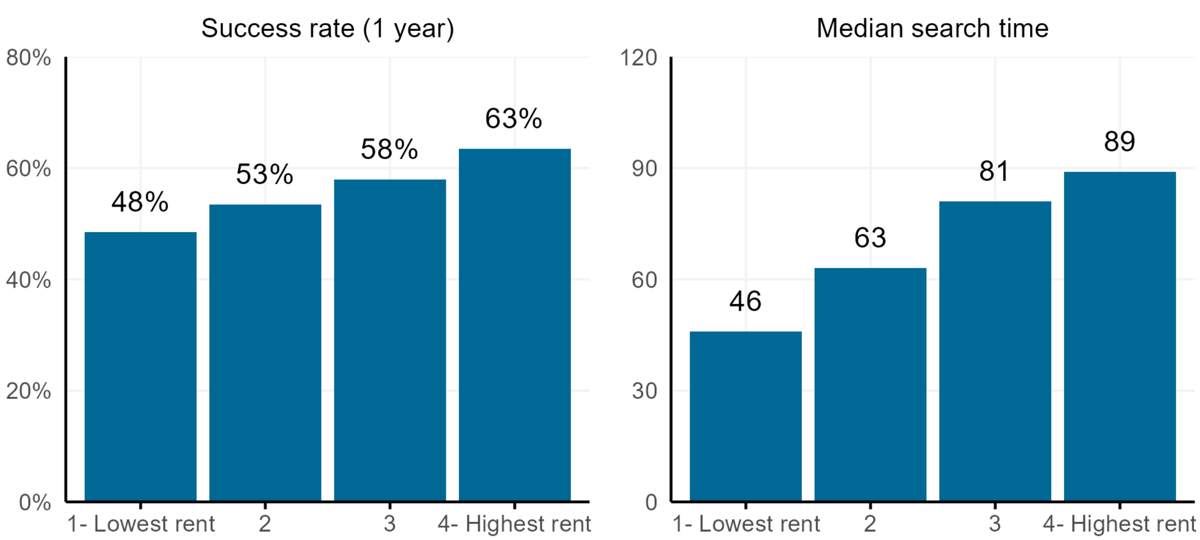
Appendix C: Variation in success rates and search times

Figure C1: Distribution of success rates and search times, 2021-2022



Note: The panel on the left shows the distribution of 1-year success rates, or the share of voucher recipients who are successful in using their voucher within one year of issuance, by PHA. The panel on the right shows the distribution of search time, or the median elapsed time between voucher issuance and successful lease-up, by PHA.

Figure C2: Success rates and search times by county median rent quartile, 2021-2022



Note: Each PHA is linked to the county where the majority of their voucher holders reside. County median rent is measured using the 5-year 2015-2019 American Community Survey.

Appendix D: Voucher issuance and lease-up across days of the month

Figure D1: Number of voucher issuance by day of month, 2015-2022

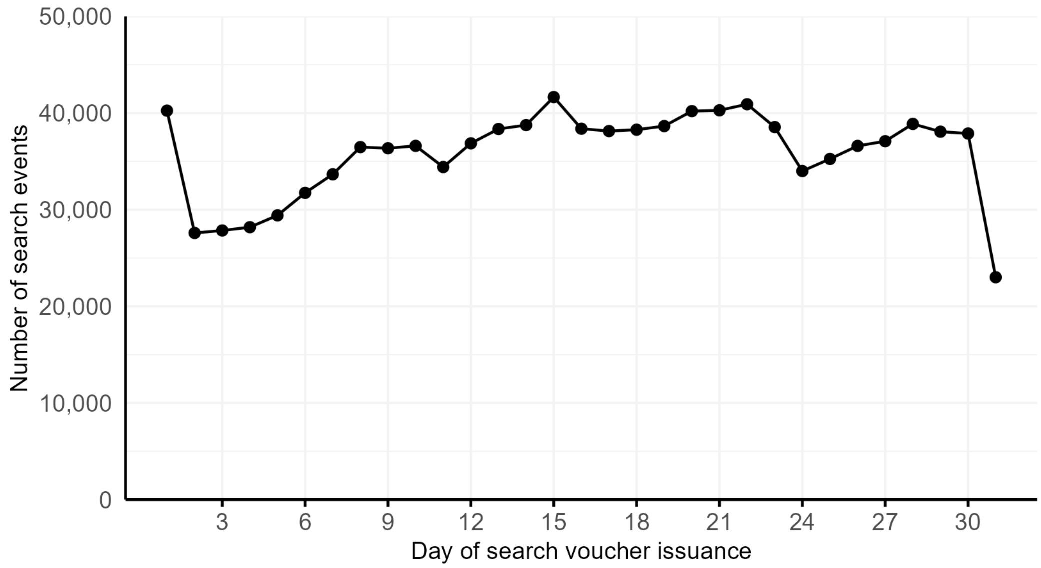


Figure D2: Share of successful searches that lease up on the first of the month, by day of voucher issuance, 2015-2022

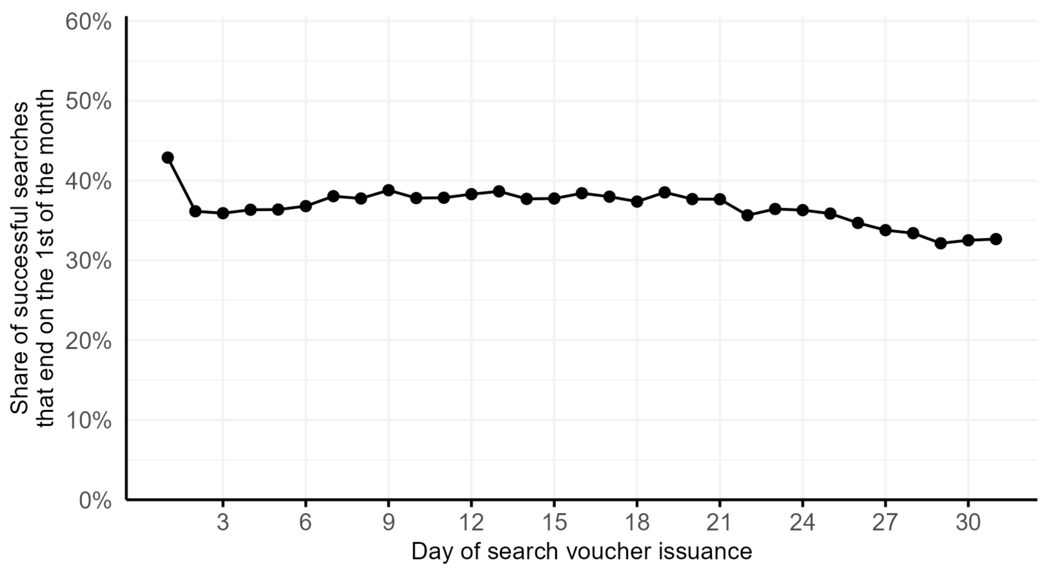
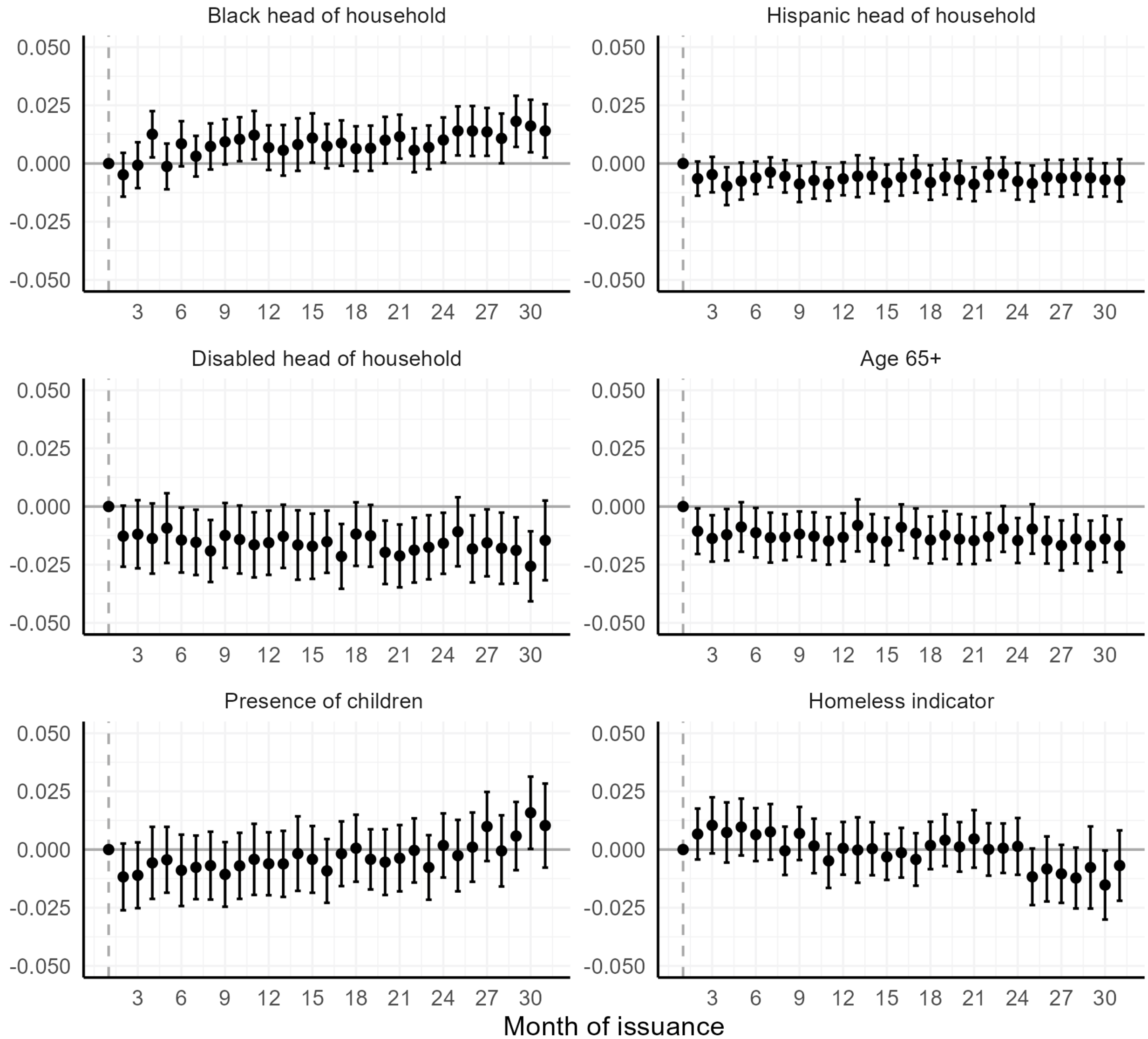


Figure D3: Balance test of individual demographic variables by day of month, 2015-2022



Note: The dependent variable in each model is an indicator for each demographic variable. Models include PHA x Year fixed effects.

Appendix E: Robustness checks

Table E1: Effect of end-of-month issuance on success rates at different search windows, with different sample restrictions

	60-day success		Unrestricted success	
	Removing disabled households	Removing special purpose vouchers	Removing disabled households	Removing special purpose vouchers
End of month indicator	-0.020***	-0.020***	0.000	0.000
	(0.002)	(0.002)	(0.002)	(0.002)
Observations	735,410	912,528	735,410	912,528
Adjusted R^2	0.109	0.108	0.089	0.089
PHA x Year fixed effects	X	X	X	X
Individual controls	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in the first two columns is an indicator for the household leasing up within 60 days; the dependent variable in the second set of columns is an indicator for the household ever leasing up. Models include individual controls (race/ethnicity, age, household composition, disability status, homeless status, citizenship status) and PHA by year fixed effects to hold allowable search time policies constant. Standard errors are clustered by PHA.

Table E2: Effect of different measures of issuance timing on success rates at different search windows

	60-day success		Unrestricted success	
	(1)	(2)	(3)	(4)
End of month indicator	-0.020***		0.000	
	(0.002)		(0.002)	
Week 2 of month		-0.006***		0.003+
		(0.002)		(0.002)
Week 3 of month		-0.012***		0.006***
		(0.002)		(0.002)
Week 4 of month		-0.025***		0.005**
		(0.002)		(0.002)
Observations	1,069,732	1,069,732	1,069,732	1,069,732
Adjusted R^2	0.106	0.106	0.083	0.083
PHA x Year fixed effects	X	X	X	X
Individual controls	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in the first two columns is an indicator for the household leasing up within 60 days; the dependent variable in the second set of columns is an indicator for the household ever leasing up. Models include individual controls (race/ethnicity, age, household composition, disability status, homeless status, citizenship status) and PHA by year fixed effects to hold allowable search time policies constant. Standard errors are clustered by PHA.

Table E3: Effect of end-of-month issuance on success rates at different search windows, controlling for conditions across months

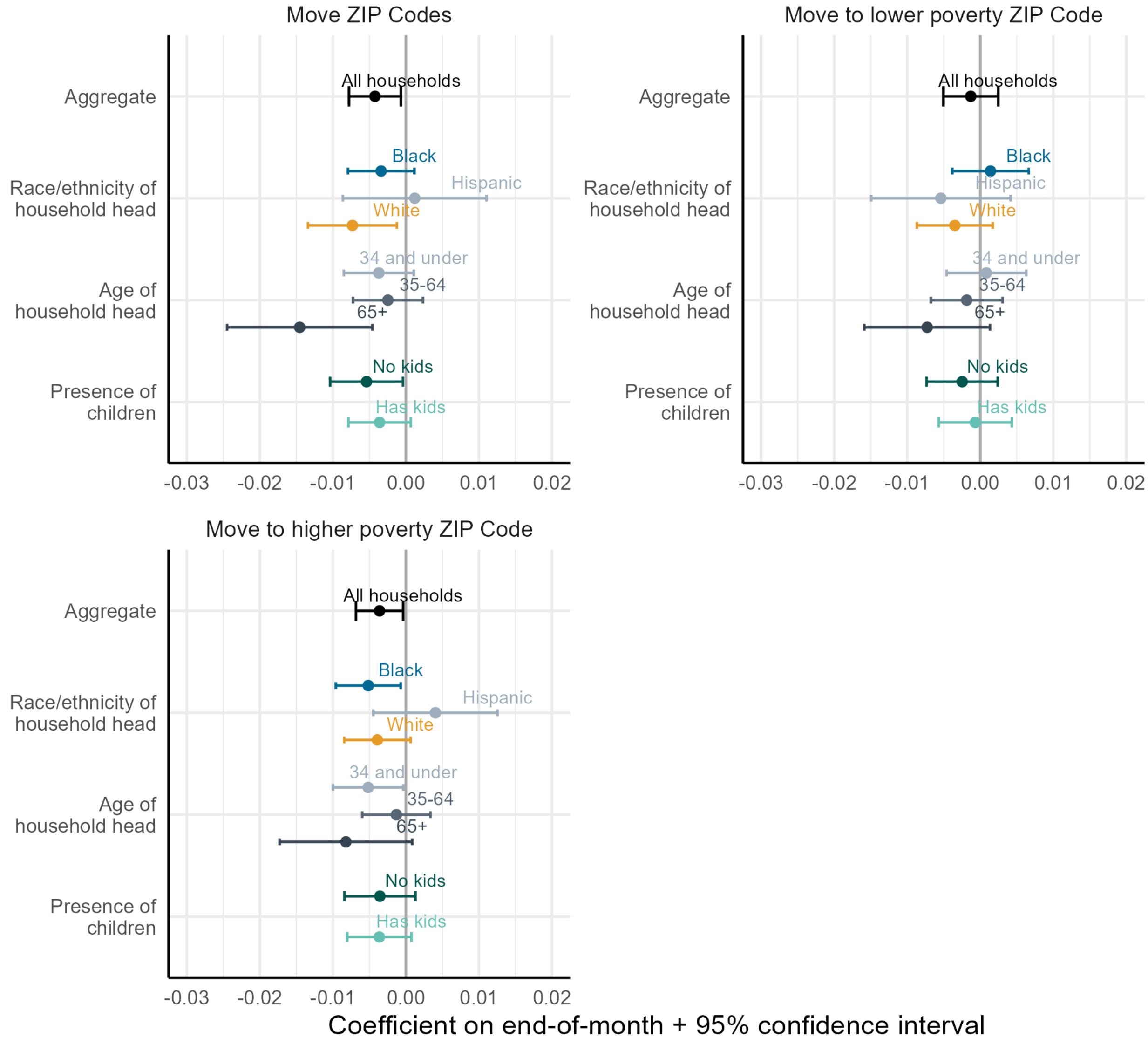
	60-day success		Unrestricted success	
	Removing December	Month fixed effects	Removing December	Month fixed effects
End of month indicator	-0.022***	-0.021***	0.000	0.000
	(0.002)	(0.002)	(0.002)	(0.002)
Observations	994,942	1,069,732	994,942	1,069,732
Adjusted R^2	0.106	0.107	0.084	0.083
PHA x Year fixed effects	X	X	X	X
Individual controls	X	X	X	X
Month fixed effects		X		X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The dependent variable in the first two columns is an indicator for the household leasing up within 60 days; the dependent variable in the second set of columns is an indicator for the household ever leasing up. Models include individual controls (race/ethnicity, age, household composition, disability status, homeless status, citizenship status) and PHA by year fixed effects to hold allowable search time policies constant. Standard errors are clustered by PHA.

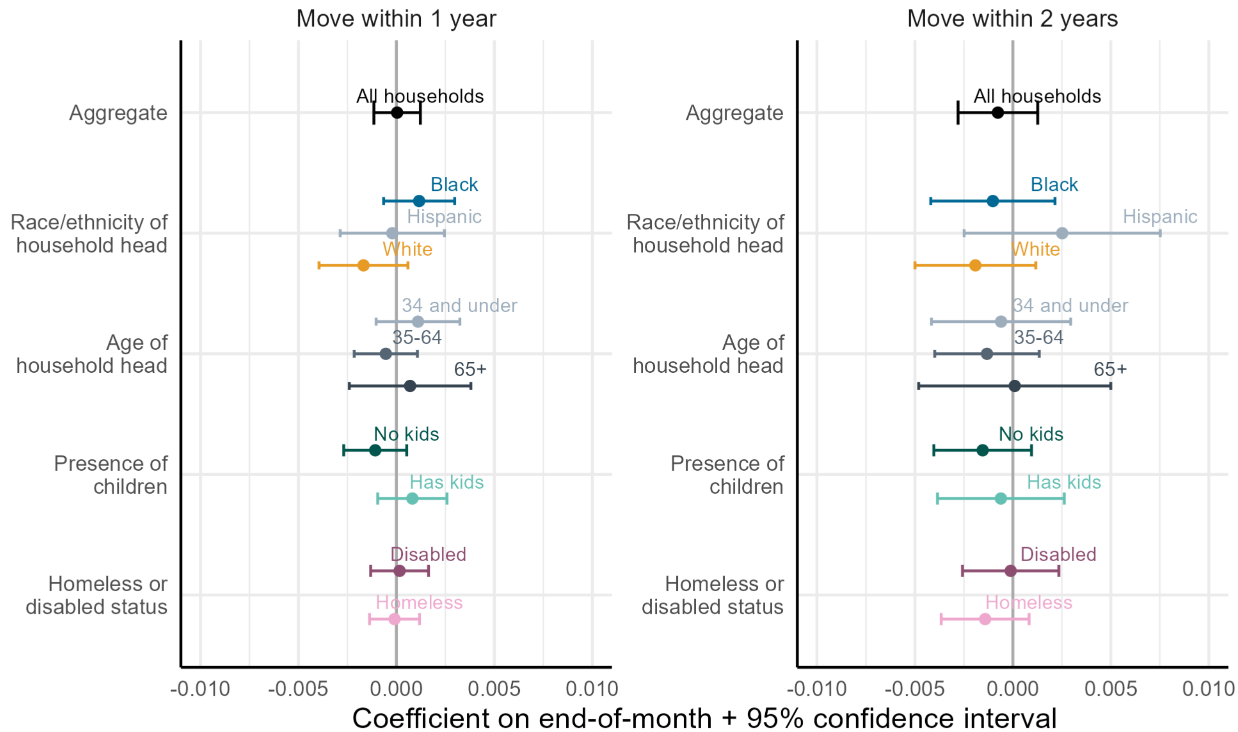
Appendix F: Heterogeneity in match quality outcomes

Figure F1: Heterogeneity in effect of end-of-month issuance on locational match quality outcomes



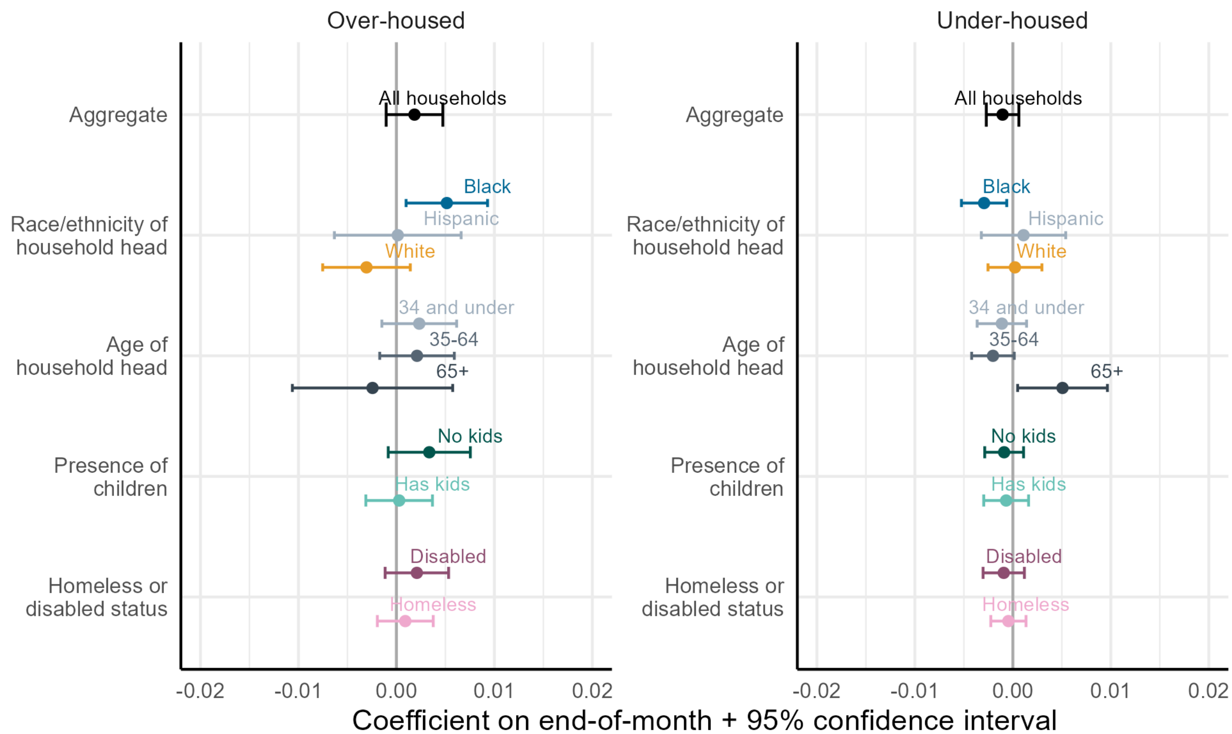
Note: Each point represents the coefficient on the end-of-month indicator variable; the dependent variable is a binary indicator for each match quality measure. Each point represents the coefficient from a stratified model. Households who are homeless at the time of issuance are excluded. Models include all other individual controls (race/ethnicity, age, household composition, disability status, citizenship status) and PHA by year fixed effect to hold allowable search time policies constant. Standard errors are clustered by PHA.

Figure F2: Heterogeneity in effect of end-of-month issuance on unit turnover match quality outcomes



Note: Each point represents the coefficient on the end-of-month indicator variable; the dependent variable is a binary indicator for each match quality measure. Each point represents the coefficient from a stratified model. Models include all other individual controls (race/ethnicity, age, household composition, disability status, homeless status, citizenship status) and PHA by year fixed effect to hold allowable search time policies constant. Standard errors are clustered by PHA.

Figure F3: Heterogeneity in effect of end-of-month issuance on relative unit size match quality outcomes



Note: Each point represents the coefficient on the end-of-month indicator variable; the dependent variable is a binary indicator for each match quality measure. Each point represents the coefficient from a stratified model. Models include all other individual controls (race/ethnicity, age, household composition, disability status, homeless status, citizenship status) and PHA by year fixed effect to hold allowable search time policies constant. Standard errors are clustered by PHA.